Algebra and Discrete Mathematics
ADM_A_B

doc. Xiaolu Hou, PhD.

FIIT, STU
xiaolu.hou @ stuba.sk

Course Outline

- Vectors and matrices
- System of linear equations
- Matrix inverse and determinants
- Vector spaces, bases and matrix transformations
- Quiz one
- Matrix transformations, fundamental spaces and decompositions
- Eulerian tours and Hamiltonian cycles
- Paths and spanning trees
- Quiz two
- Trees and networks
- Matching

Recommended reading

- Saoub, K. R. (2017). A tour through graph theory. Chapman and Hall/CRC.
  - Sections 1.1 – 1.5, 2.1 – 2.3
  - Accessible online (free copy)
  - Alternative download link

Lecture outline

- Definitions and terminologies
- Different types of graphs
- Touring a graph
- Eulerian circuit algorithms
- Eulerization
- Existence of Hamiltonian cycles
- Traveling salesman problem
- Digraphs and Asymmetric Traveling Salesman Problem

Eulerian tours and Hamiltonian cycles

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Graph

Definition

A *graph* consists of two sets: $V(G)$, called the *vertex set*, and $E(G)$, called the edge set. An *edge*, denoted xy, is an unordered pair of vertices $x, y \in V(G)$.

- We will often use G or $G = (V, E)$ as short-hand.
- xy and yx are treated equally, though it is customary to write them in alphabetical order
- Later in the course: *directed graphs* – the order in which an edge is written provides additional meaning.

Graph – example

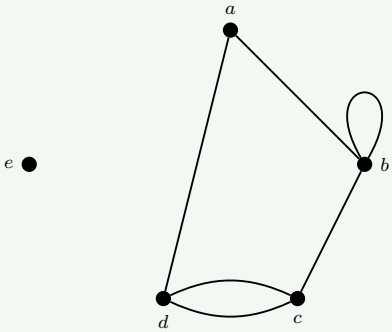
Example

- Let G be a graph where

$$V(G) = \{a, b, c, d, e\},$$

$$E(G) = \{ab, ad, bc, cd, cd, bb\}$$

- Visualization of G: a dot represents a vertex and an edge is a line connecting the two dots (vertices)
- Note: two edges between vertices c and d; a loop at b; no edges at e



Terminologies

Definition

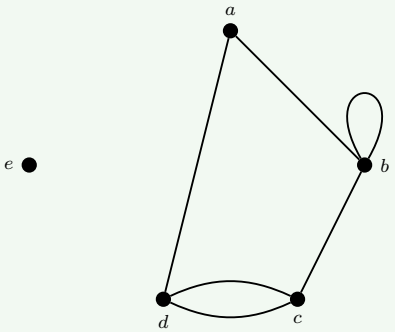
Let $G = (V, E)$ be a graph

- If $xy \in E$, then x and y are the *endpoints* of the edge xy. Each of x and y is *incident to* the edge xy.
- *Adjacent* edges: share an endpoint
- *Adjacent* vertices (*neighbors*): incident to the same edge
- $N(v)$: the set of all neighbors of a vertex v
- *Isolated vertex*: not incident to any edge
- *Loop*: both endpoints of an edge are the same vertex
- *Multi-edges*: two or more edges with the same endpoints
- *Degree* of a vertex: number of edges incident to the vertex, where a loop contributes 2. Denoted $\deg(v)$. If $\deg(v)$ is even (resp. odd), then v is called an *even* (resp. *odd*) vertex.

Graph – example

Example

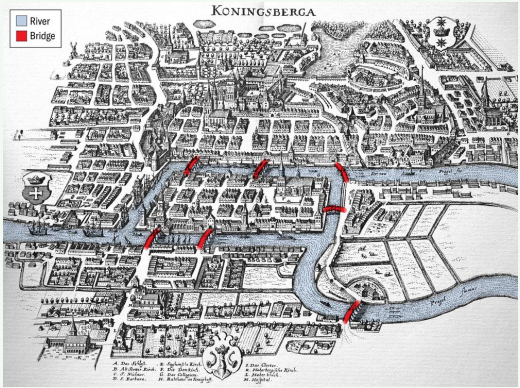
- Edges ab and ad are adjacent
- a and b are adjacent vertices
- $N(d) = \{a, c\}$, $N(b) = \{a, b, c\}$
- e is an isolated vertex
- bb is a loop
- cd is a multi-edge
- $\deg(a) = 2$, $\deg(b) = 4$, $\deg(c) = 3$, $\deg(d) = 3$, $\deg(e) = 0$



Modeling with a graph – Königsberg bridge problem

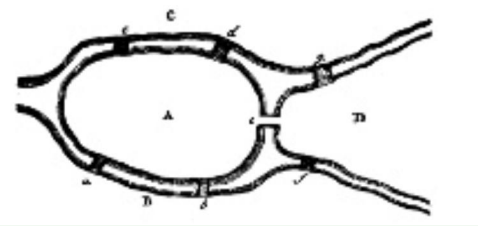
Example

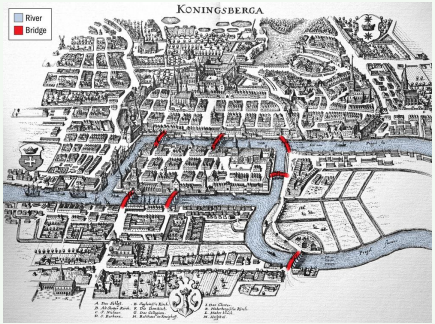
- In 1736, Leonhard Euler published a short paper on the bridges of Königsberg.
- Can you leave your home, travel across each of the bridges in the city exactly once, and then return home?
- The puzzle is described as the birth of graph theory



Modeling with a graph – Königsberg bridge problem

Example

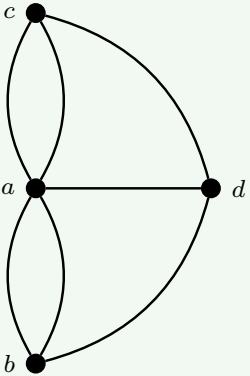


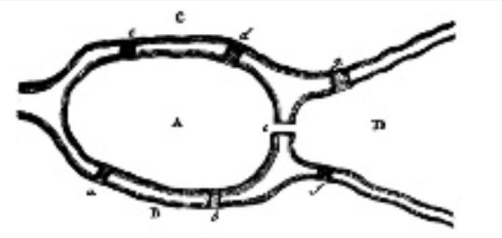


Euler reduced the map of Königsberg to a simpler version where only the relationships between landmasses were of importance

Modeling with a graph – Königsberg bridge problem

Example





We can use a simpler way to model the city with a graph – the vertices represent an island, a south bank, a north bank, and a peninsula (a, b, c, d respectively) – can you leave your home (vertex), travel across each of the bridges (edges) in the city (graph) exactly once and then return home?

Order and size

Definition

- A graph $G = (V(G), E(G))$ is called *finite* if both $V(G)$ and $E(G)$ are finite.
- A graph that is not finite is called an *infinite* graph
- The *order* of G is the number of vertices of G, i.e. $|V(G)|$.
- The *size* of G is the number of edges of G, i.e. $|E(G)|$.

Eulerian tours and Hamiltonian cycles

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Simple graph

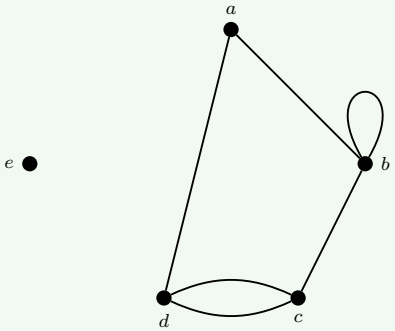
Definition

- *Pseudograph*: can contain multi-edges and loops
- *Multigraph*: can contain multi-edges, but no loops
- *Simple graph*: no multi-edges or loops
- *Trivial graph*: one vertex, no edges

Graphs – example

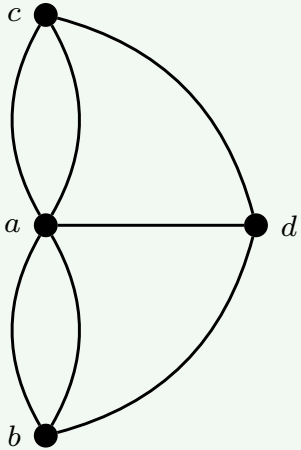
Example

Pseudograph



Order: 5, size: 6

Multigraph

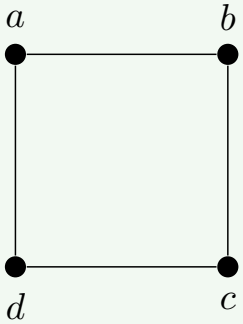


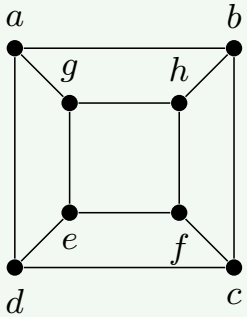
Regular graph

Definition

- *Regular graph*: all of its vertices have the same degree
- k-*regular*: all vertices have degree k
- A 3-regular graph is also called a *cubic* graph

Example





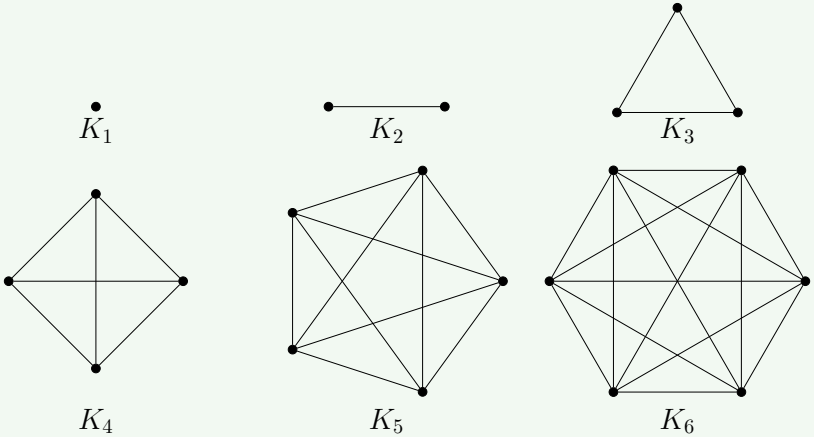
Complete graph

Definition

A simple graph G is *complete* if every pair of distinct vertices is adjacent. The complete graph on n vertices is denoted K_n.

The first six complete graphs

Example



Complement of a graph

Definition

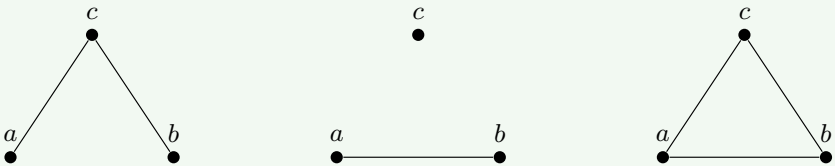
Let $G = (V, E)$ be a simple graph. The *complement* of G, denoted $\overline{G} = (V, \overline{E})$, is the graph with the same vertex set as G, where

$$uv \in \overline{E} \quad \text{if and only if} \quad uv \notin E$$

for all distinct vertices $u, v \in V$.

- G and $\overline{G}$ together forms a complete graph

Example



Bipartite graphs

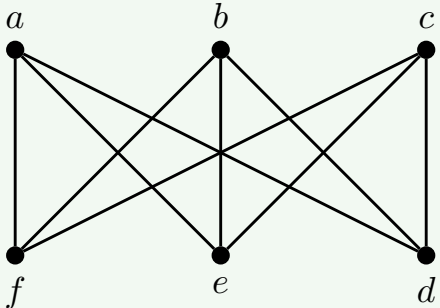
Definition

A graph $G = (V, E)$ is *bipartite* if the vertices can be partitioned into two sets, V_1 and V_2, so that

$$V_1 \cap V_2 = \emptyset, \quad V = V_1 \cup V_2,$$

and every edge has exactly one endpoint in V_1 and the other endpoint in V_2.

Example



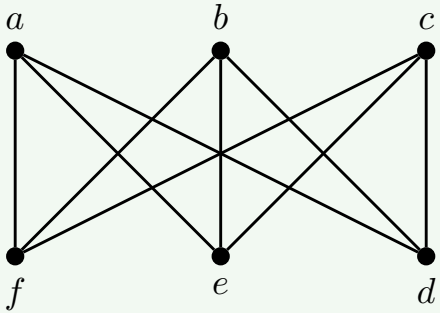
Complete bipartite graph

Definition

A simple bipartite graph $G = (V_1 \cup V_2, E)$ is a *complete bipartite graph* if every vertex in V_1 is adjacent to every vertex in V_2. If $|V_1| = m$, $|V_2| = n$, we write $K_{m,n}$

Example

$K_{3,3}$



Eulerian tours and Hamiltonian cycles

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Definitions

Definition

Let G be a graph

- *Walk*: a sequence of vertices so that there is an edge between consecutive vertices. A walk can repeat vertices and edges.
- *Trail*: a walk with no repeated edges
- *Path*: a trail with no repeated vertices. A path on n vertices is denoted P_n.
- *Closed walk*: a walk that starts and ends at the same vertex
- *Circuit*: a closed trail
- *Cycle*: a closed path. A cycle on n vertices is denoted C_n.

Length of any of these tours is the number of edges.

- Path $\subseteq$ Trail $\subseteq$ Walk
- Cycle $\subseteq$ Circuit $\subseteq$ Closed walk

Length of a tour – example

Example

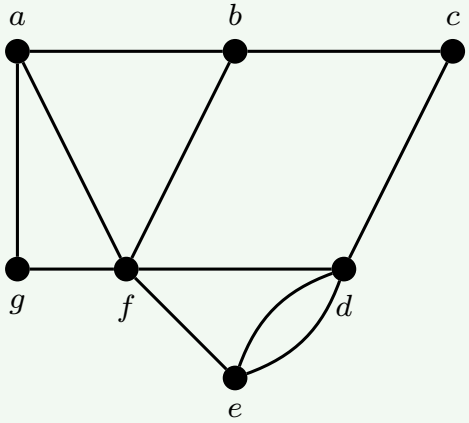
- P_n has length $n-1$
- C_n has length n

Touring a graph – example

Example

Given the graph, find

- a trail (that is not a path) from a to c
- a path from a to c
- a circuit (that is not a cycle) starting at b
- a cycle starting at b



Touring a graph – example

Example

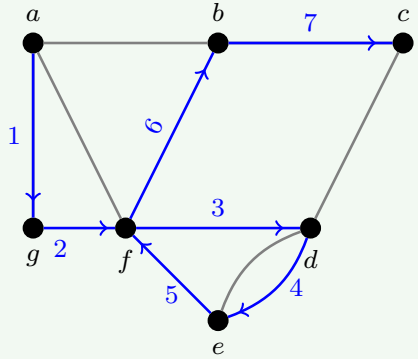


Figure: A trail from a to c

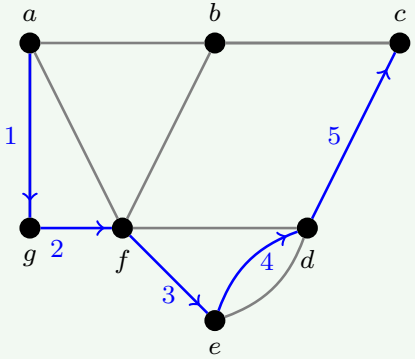


Figure: A path from a to c

Touring a graph – example

Example

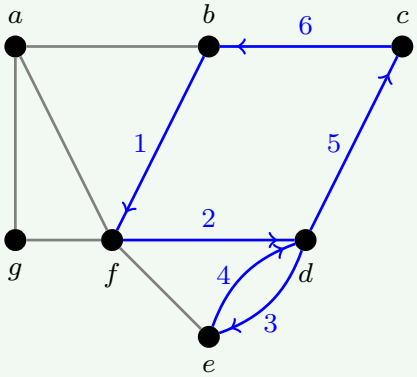


Figure: A circuit starting at b

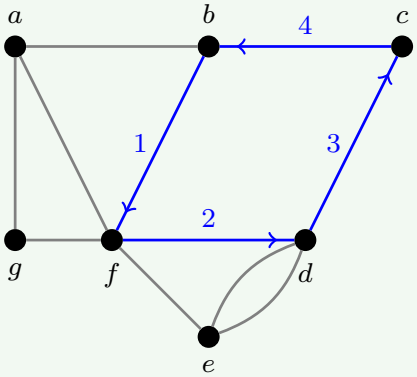


Figure: A cycle starting at b

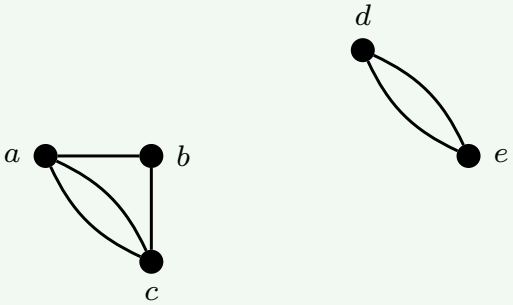
Connected vertices

Definition

Let G be a graph. Two vertices x and y are *connected* if there exists a path from x to y in G. The graph G is *connected* if every pair of distinct vertices is connected.

Example

Not connected



Eulerian circuit

Definition

Let G be a graph. An *Eulerian circuit* is a circuit that uses every edge of G exactly once. An *Eulerian trail* is a trail that uses every edge of G exactly once. If G contains an Eulerian circuit, then G is called *Eulerian*. If G contains an Eulerian trail but not an Eulerian circuit, then G is called *semi-Eulerian*.

Eulerian graph

Theorem

A graph G is Eulerian if and only if

- *G is connected and*
- *every vertex has an even degree*

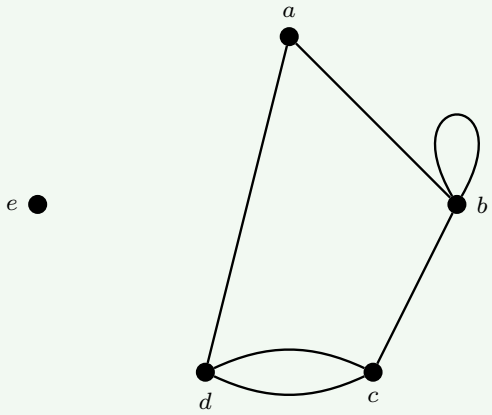
A graph G is semi-Eulerian if and only if

- *G is connected and*
- *exactly two vertices have odd degree*

- When traveling through a graph, we need to pair each entry edge with an exit edge.
- If a vertex is odd, then there is no pairing available and we would eventually get stuck at that vertex
- If the starting and ending locations are different, then exactly two vertices must be odd since the first edge out of the starting vertex does not need to be paired with a return edge and the last edge to the ending vertex does not need to be paired with an exit edge

Eulerian and semi-Eulerian graphs – example

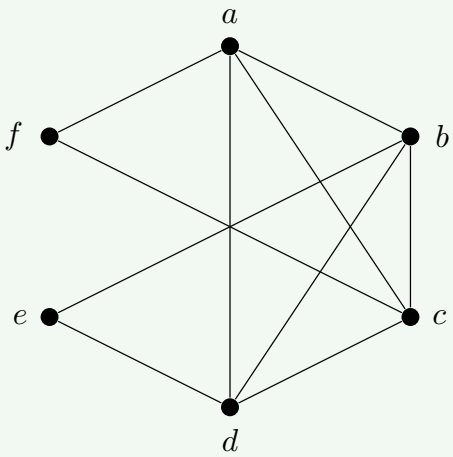
Example



Not connected – not Eulerian, not semi-Eulerian. The graph has two *components*.

Eulerian and semi-Eulerian graphs – example

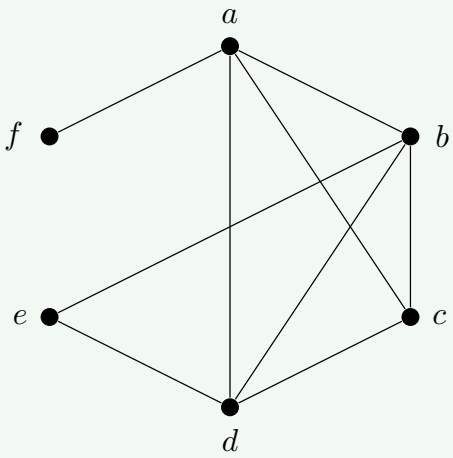
Example



Eulerian – all vertices have degree 2 or 4

Eulerian and semi-Eulerian graphs – example

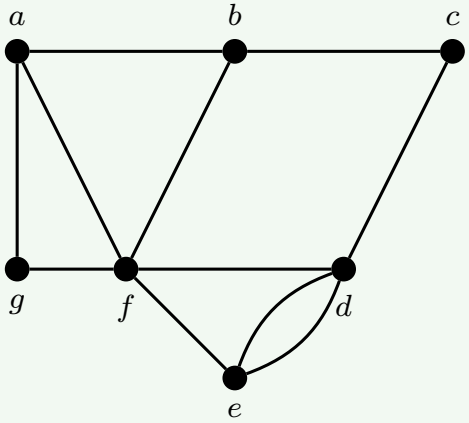
Example



semi-Eulerian – exactly two vertices are odd

Eulerian and semi-Eulerian graphs – example

Example



Connected, but not Eulerian or semi-Eulerian - more than two odd vertices (a, b, e, f)

Handshaking Lemma

Theorem (Handshaking Lemma)

Let $G = (V, E)$ be a graph. Suppose $V = \{v_1, v_2, \ldots, v_n\}$, then

$$\deg(v_1) + \deg(v_2) + \cdots + \deg(v_n) = 2|E|$$

Proof.

Since each edge contributes two to the sum of degrees, one at each of the two endpoints, the sum of degrees is even and equal to twice the number of edges. □

Corollary

Corollary

In any graph, there is an even number of odd vertices.

Proof.

$$\sum_{v_i \text{ has odd degree}} \deg(v_i) + \sum_{v_j \text{ has even degree}} \deg(v_j) = 2|E|$$

The second term on the left is even, the sum is even. □

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Fleury's Algorithm

- Find an Eulerian circuit or an Eulerian trail
- By the previous theorem, the input graph must be connected and has zero or two odd vertices

Fleury's Algorithm

1. Choose a starting vertex:
   - If the graph G has no vertices of odd degree, start from any vertex
   - If G has exactly two vertices of odd degree, start from one of them
2. Choose an edge to traverse: Select an edge incident to the current vertex v, ensuring the graph containing remaining edges is connected after removing it
3. Traverse and remove the edge: Move to the other endpoint of the selected edge, then delete the edge from the graph. The new vertex becomes the current vertex.
4. Repeat steps 2 and 3 until there are no more edges left in the graph

Output

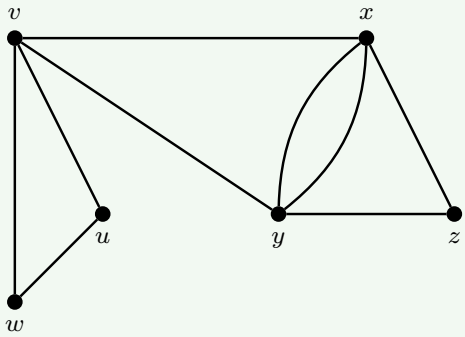
- If G originally had no odd-degree vertices, Eulerian circuit
- If G had exactly two odd-degree vertices, Eulerian trail

Practical Implementation: Maintain two copies of the graph:

- One for tracking the Eulerian path or cycle.
- Another for dynamically removing edges

Fleury's Algorithm – example

Example



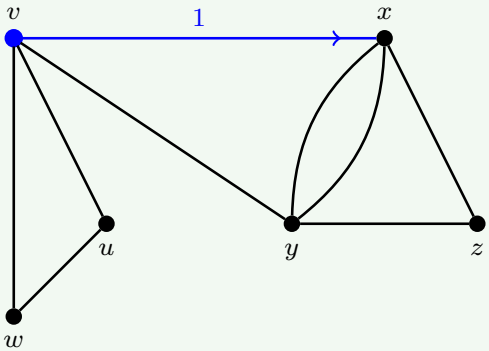
The graph is Eulerian:

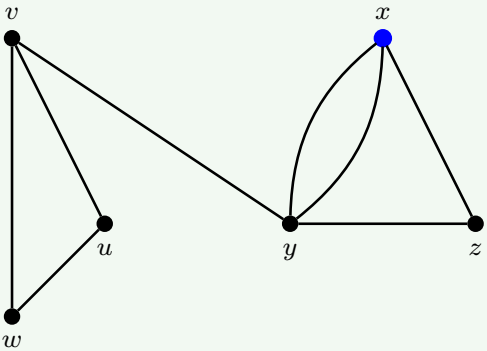
$\deg(u) = 2, \quad \deg(v) = 4, \quad \deg(w) = 2, \quad \deg(x) = 4, \quad \deg(y) = 4, \quad \deg(z) = 2$

- Step 1. start vertex v
- Step 2. choose edge vx

Fleury's Algorithm – example

Example

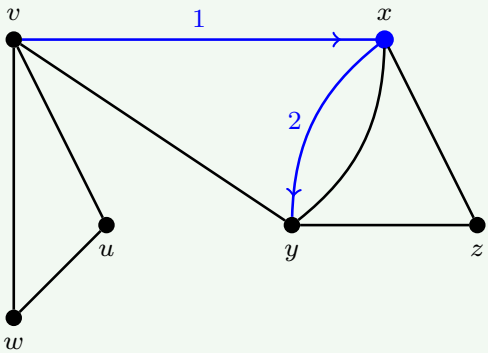


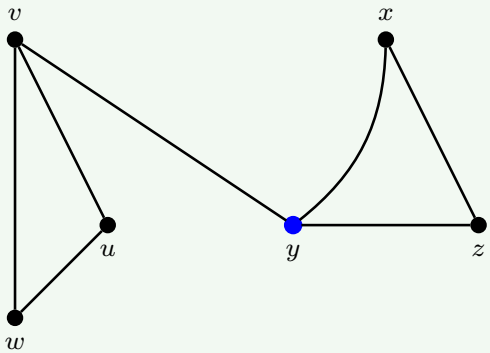


- Step 3. travel to vertex x and delete edge vx. Current vertex: x.

Fleury's Algorithm – example

Example

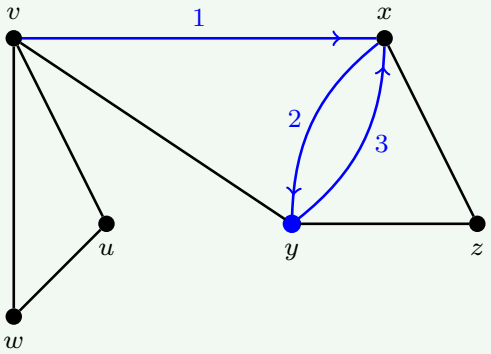


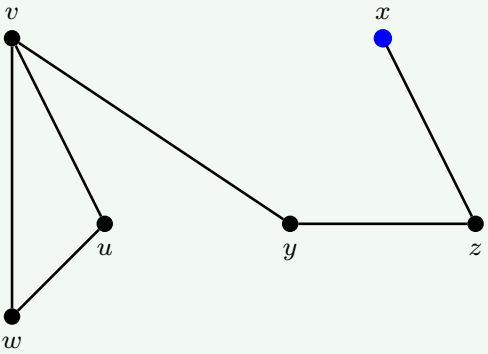


- Step 2. choose edge xy
- Step 3. travel to vertex y and delete edge xy. Current vertex: y.

Fleury's Algorithm – example

Example

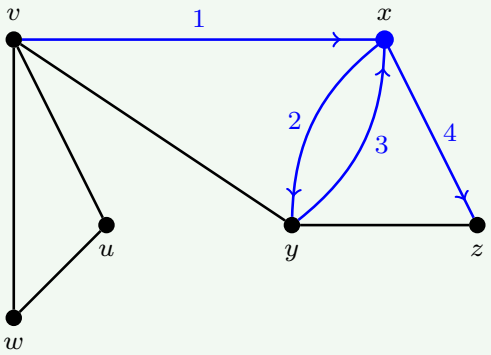


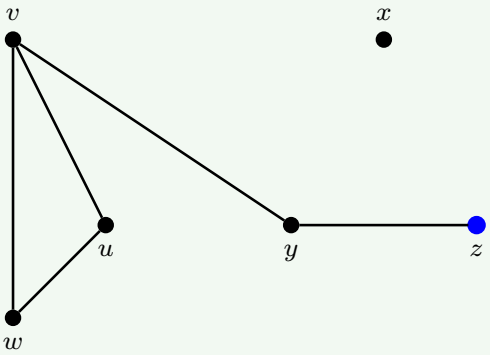


- Step 2. choose edge yx, note: cannot choose edge yv, its removal would disconnect the remaining graph
- Step 3. travel to vertex x and delete edge yx. Current vertex: x.

Fleury's Algorithm – example

Example

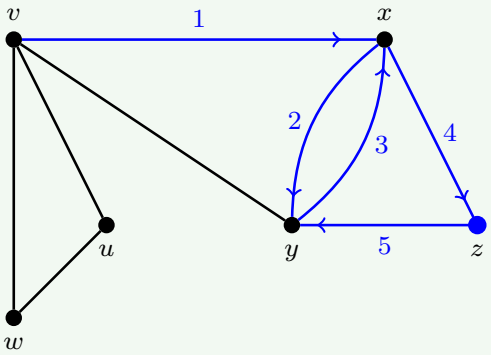


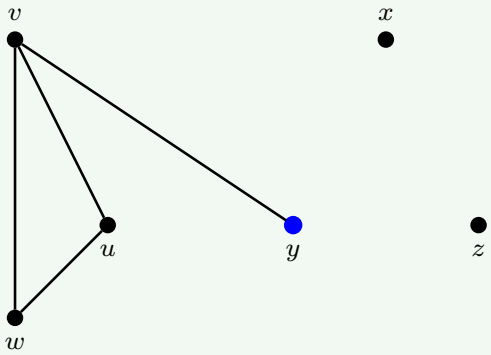


- Step 2. only one available edge xz
- Step 3. travel to vertex z and delete edge xz. Current vertex: z.

Fleury's Algorithm – example

Example

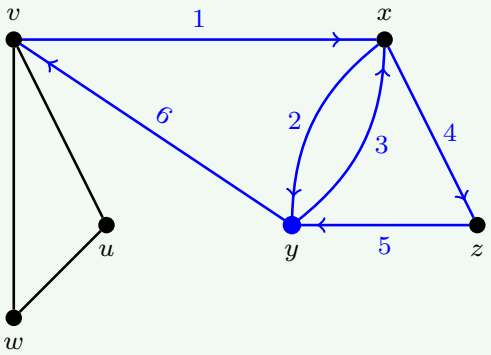


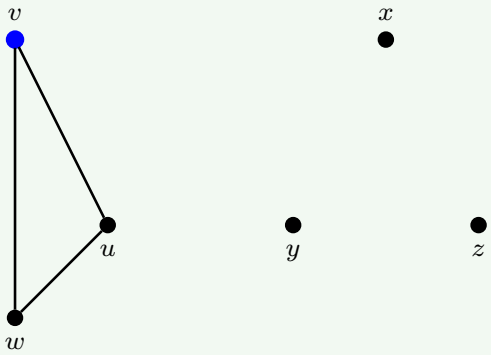


- Step 2. only one available edge zy
- Step 3. travel to vertex y and delete edge zy. Current vertex: y.

Fleury's Algorithm – example

Example

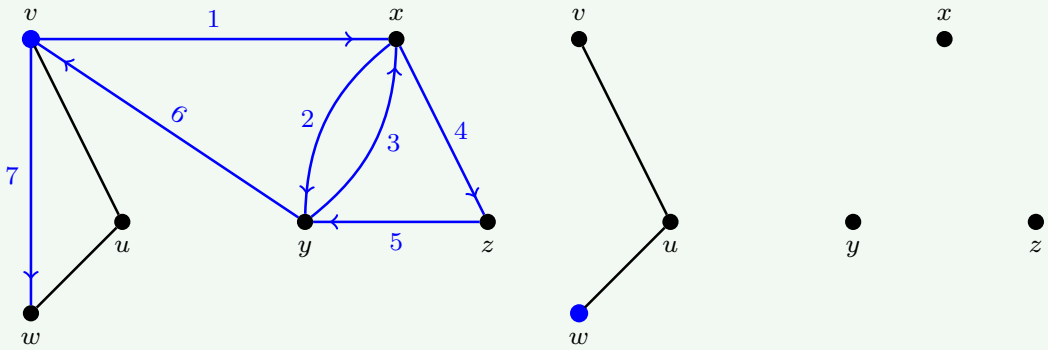




- Step 2. only one available edge yv
- Step 3. travel to vertex v and delete edge yv. Current vertex: v.

Fleury's Algorithm – example

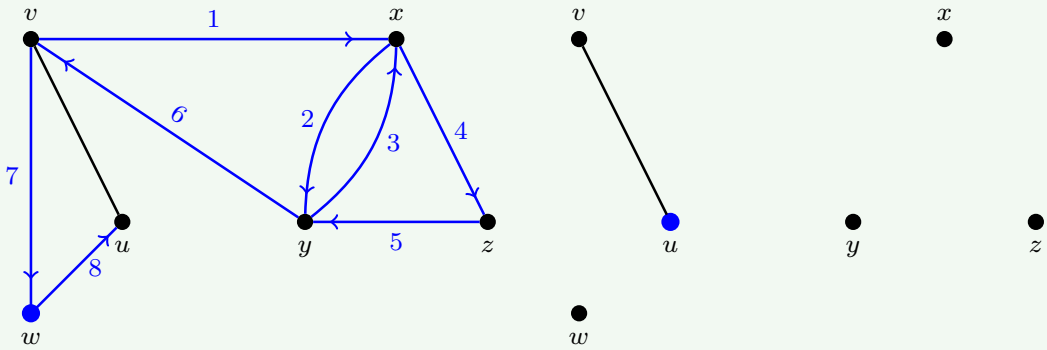
Example



- Step 2. both vu and vw can be chosen. Here we choose vw.
- Step 3. travel to vertex w and delete edge vw. Current vertex: w.

Fleury's Algorithm – example

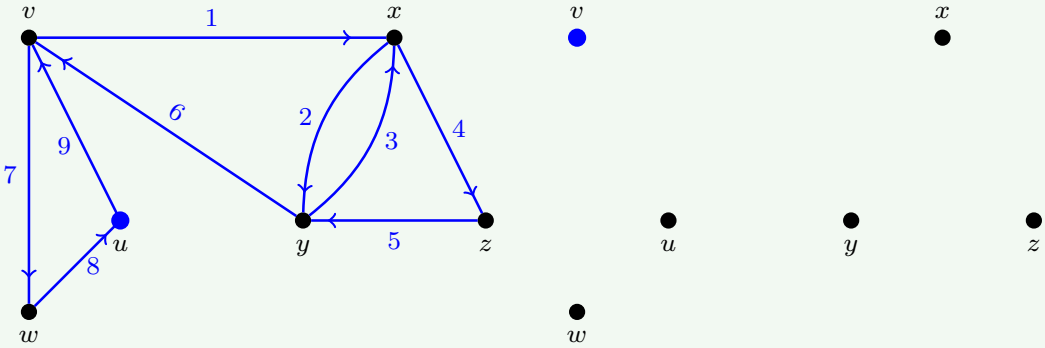
Example



- Step 2. only one available edge wu
- Step 3. travel to vertex u and delete edge wu. Current vertex: u.

Fleury's Algorithm – example

Example



- Step 2. only one available edge uv
- Step 3. travel to vertex v and delete edge uv
- We have obtained an Eulerian circuit starting and ending at vertex v

Hierholzer's Algorithm

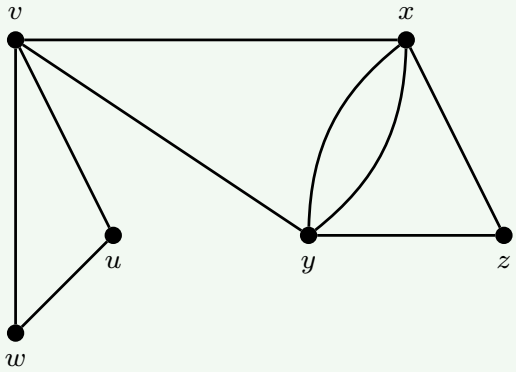
- Find an Eulerian circuit
- Input: an Eulerian graph G – connected, all vertices are even

Hierholzer's Algorithm

1. Choose a starting vertex, say v, and find an initial circuit C starting at v.
2. If there exists a vertex x in the current circuit C that has unused edges, form a new circuit C' starting at x that uses two of those unused edges
3. Merge the newly found circuit C' into the existing circuit C, updating C accordingly.
4. Repeat steps 2 and 3 until all edges of G have been traversed.

Hierholzer's Algorithm – example

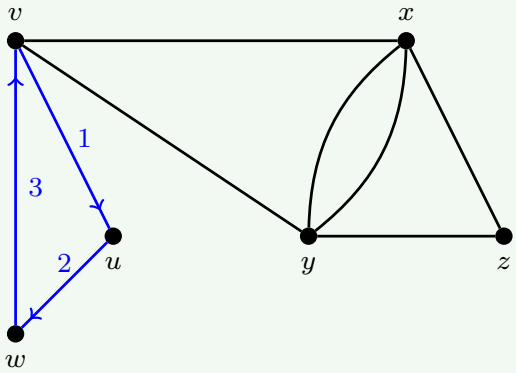
Example



Same graph, we have discussed that the graph is Eulerian.

Hierholzer's Algorithm – example

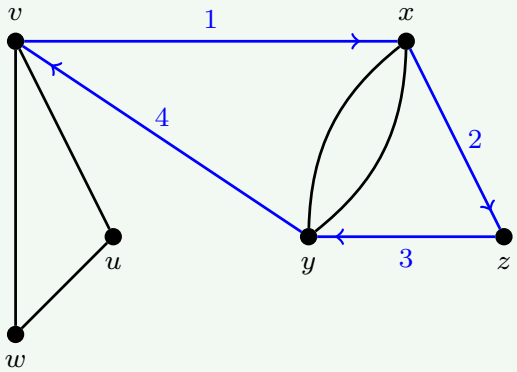
Example



- Step 1: choose vertex v and find a circuit starting at v

Hierholzer's Algorithm – example

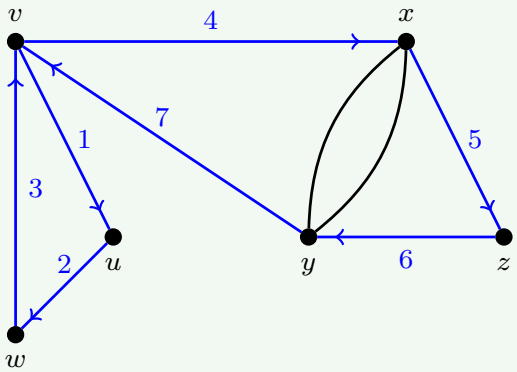
Example



- Step 2: since $\deg(v) = 4$, two unused edges remain incident to v. We can find a second circuit starting at v.

Hierholzer's Algorithm – example

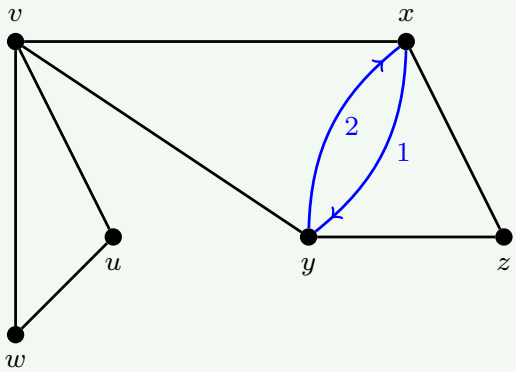
Example



- Step 3: combine the two circuits from Step 1 and Step 2. There are multiple ways to combine two circuits.

Hierholzer's Algorithm – example

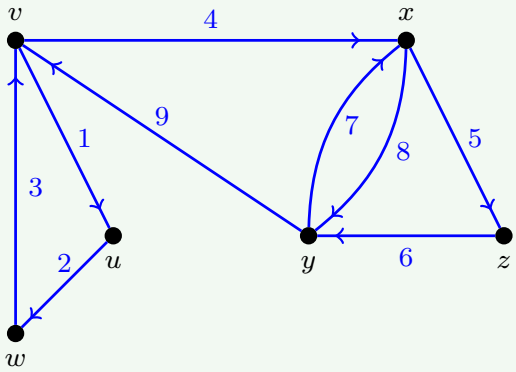
Example



- Step 2: $\deg(x) = 4$ and two edges remain for x, we can find a circuit starting from x

Hierholzer's Algorithm – example

Example



- Step 3: combine the new circuit with the existing one
- We have obtained an Eulerian circuit

Eulerian tours and Hamiltonian cycles

- Definitions and terminologies
- Different types of graphs
- Touring a graph
- Eulerian circuit algorithms
- Eulerization
- Existence of Hamiltonian cycles
- Traveling salesman problem
- Digraphs and Asymmetric Traveling Salesman Problem

Eulerization – definition

Definition

Given a connected graph $G = (V, E)$, an *Eulerization* of G is the graph $G' = (V, E')$ so that

- G' is obtained by duplicating edges of G, and
- every vertex of G' is even

A *semi-Eulerization* of G results in a graph G' so that

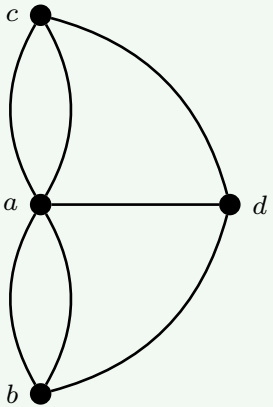
- G' is obtained by duplicating edges of G, and
- exactly two vertices of G' are odd

Optimal exhaustive tour

- In the context of the Königsberg Bridge Problem, duplicating an edge would be equivalent to walking the same bridge twice
- Although this solution would be outside the original parameters (walking each bridge exactly once), it allows for an approximate solution using the fewest number of duplications.
- This is referred to as finding an *optimal exhaustive tour* of a graph

The Königsberg Bridge Problem

Example

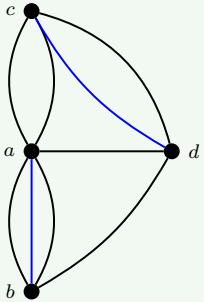


We have discussed that the graph modeling Königsberg city is not Eulerian or semi-Eulerian

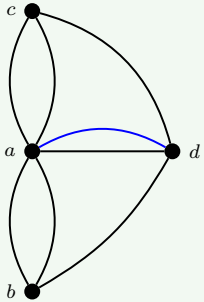
The Königsberg Bridge problem

Example

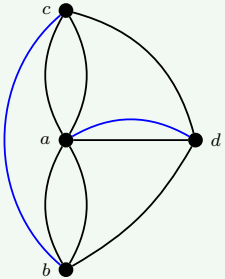
Eulerization



semi-Eulerization



not an Eulerization



- Note that when creating the new graph, edges must be duplicated, not added
- Of course edges can be added in a city, but unrealistic from the standpoint of a person touring the bridges of a city during a specific moment in time. Hence, we only allow duplications of edges

The optimization question

- We are interested in the question: how can we Eulerize (or semi-Eulerize) a graph using the fewest number of edge duplications?
- We are attempting to find an optimal tour of the graph, that is minimize the total length of the circuit (or trail).

Eulerization method

1. Identify the odd vertices of the graph
2. Pair up the odd vertices, trying to pair as many adjacent vertices as possible while also avoiding paring vertices far away from each other[1]
3. Duplicate the edges along an optimal path from one vertex to its pair

Note:

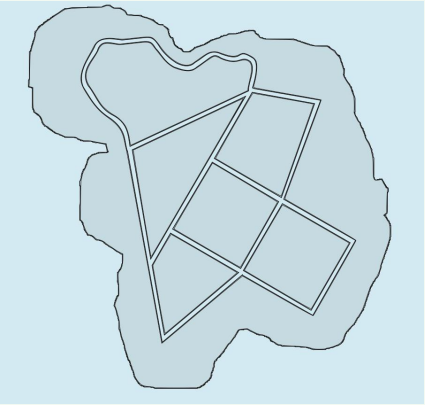
- In the process of determining which edges to duplicate along optimal paths, never repeat an edge more than once
- If an edge is crossed three times, removing two of the duplications will not change the parity of the endpoint of the edge
  - that is, a vertex will remain odd or remain even when subtracting two from the degree

[1] Distance: the length of the shortest path connecting those two vertices.

Eulerization – example

Example

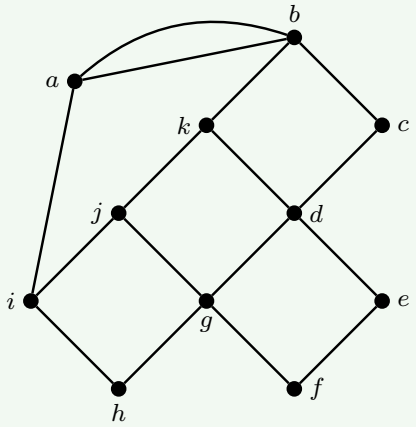
- The citizens of the small island town of Sunset Island want to hire a night patrol during the busy summer tourist season
- Model the town as a graph and find an optimal Eulerization of the graph

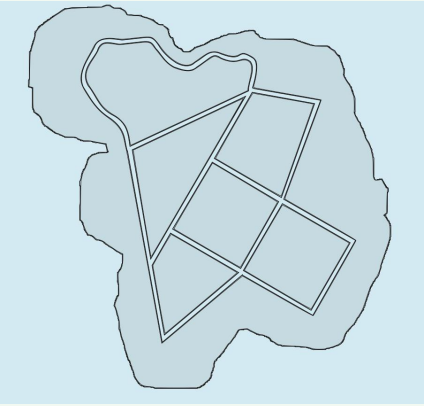


Eulerization – example

Example

The graph modeling - vertices represent intersections and edges represent street blocks

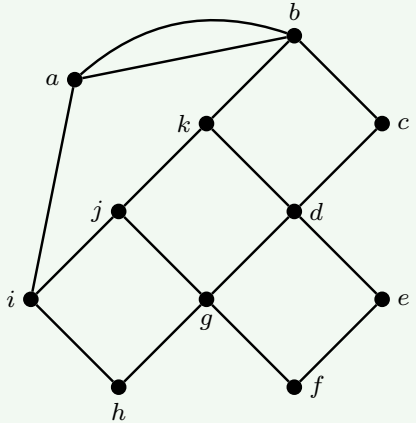


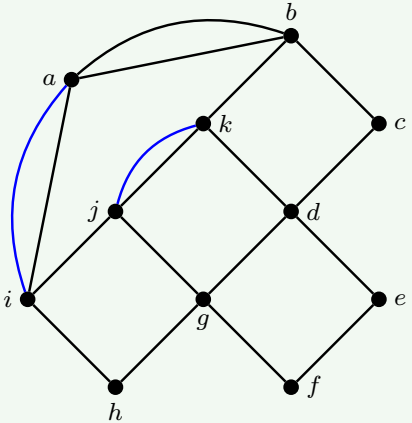


Eulerization – example

Example

- Four odd vertices: a, i, j, k
- They can be split into two pairs of adjacent vertices: a, i and j, k

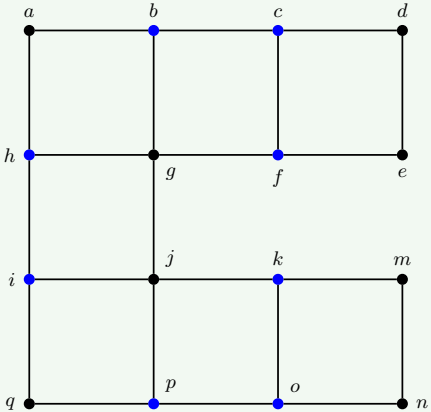


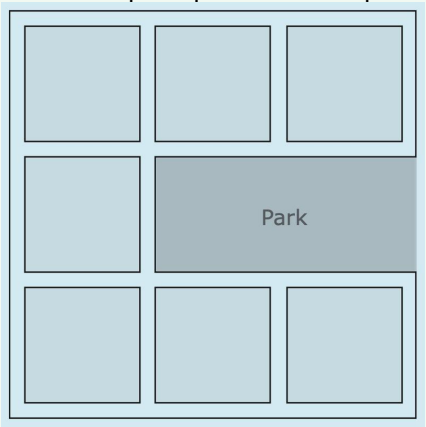


Eulerization – example

Example

- Task: put up fliers along each block of a small portion of a town
- Requirement: travel along each street once but cannot put up fliers in the park



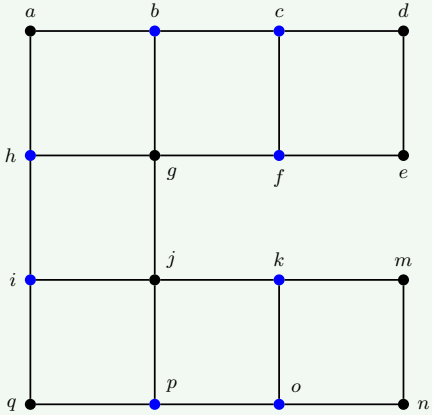


We model the graph and identify the odd vertices (in blue)

Eulerization – example

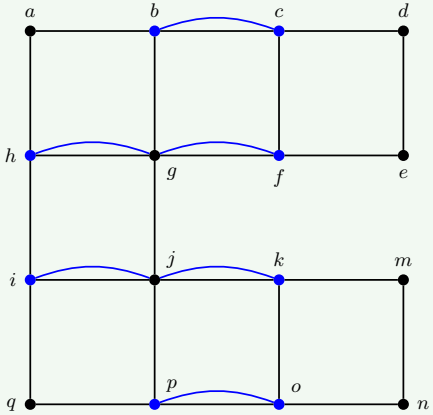
Example

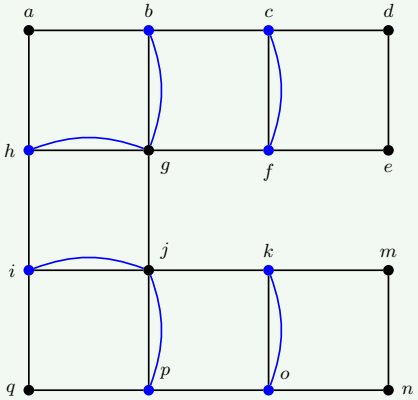
- Unlike the previous example, we cannot split all the 8 odd vertices into adjacent pairs, which corresponds to only 4 duplicated edges
- 5 duplicated edges is also not possible
  - if we use exactly 2 adjacent pairs, we are left with 2 pairs each of which is of distance 2 apart and thus requiring another 4 edges duplications
  - if we use 3 adjacent pairs, we are left with 1 pair of distance 3 apart, which requires 3 more edges
- At least 6 edge duplications are needed



Eulerization – example

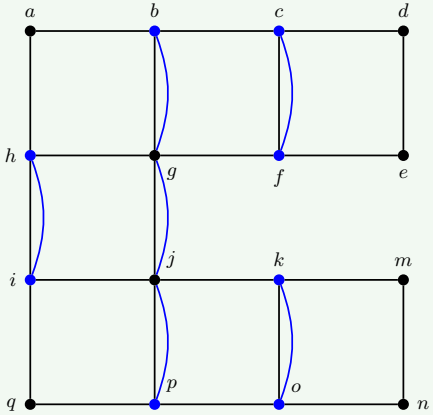
Example

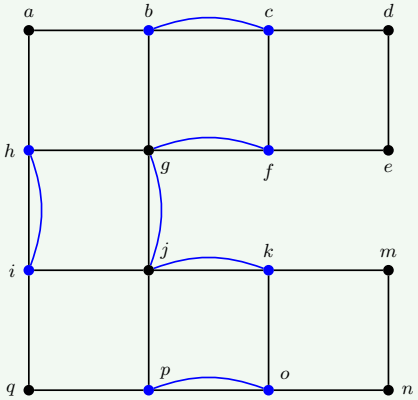




Eulerization – example

Example





Weighted graph

- In the previous examples, the streets follow a highly structured grid layout, where traveling along one block is comparable to traveling along another.
- What happens when different streets require varying levels of effort to traverse?

Definition

A *weighted graph* $G = (V, E, \omega)$ is a graph where each of the edges has a real number associated with it. This number is referred to as the *weight* and denoted $\omega(xy)$ for an edge xy

Remark

- Weight of an edge can represent length, time, cost, etc.
- A weighted graph can also refer to a graph in which each of the vertices is assigned a weight

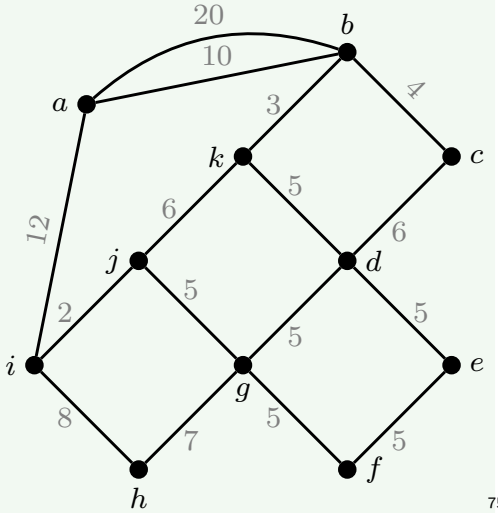
Chinese Postman Problem

- The weighted version of an Eulerization problem is called the *Chinese Postman Problem*
- The name originates not from anything particular about postmen in China, but rather from the mathematician who first proposed the problem —the Chinese mathematician Mei-Ku Kwan (管梅谷) in 1962

Chinese Postman Problem – example

Example

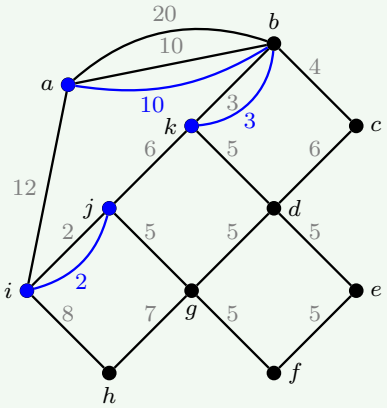
- Suppose the travel time through the streets of Sunset Island varies depending on the street.
- Find an optimal Eulerization taking into account the weights

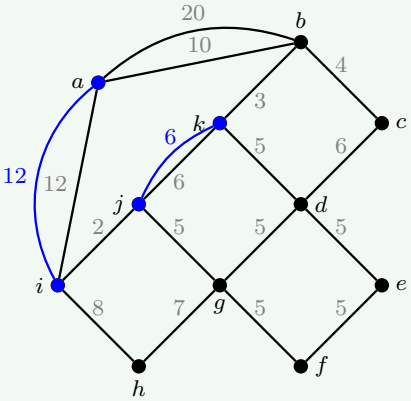


Chinese Postman Problem – example

Example

The previous Eulerization we have obtained (on the right) is not optimal (two duplicated edges total 18). A better Eulerization duplicates three edges for a total of 15





Remarks

- Be careful when duplicating edges in a graph with multi-edges, indicating which edge has been duplicated – including the edge weight clarifies which option was used.
- In general, solving the Chinese Postman Problem can be quite challenging
- Most small examples can be solved by inspection
- The choice of which edges to duplicate when working with a weighted graph relies in part on shortest paths between two vertices.
- The difficulty is in choosing which vertices to pair.
- This will be discussed later in the course

About tutorials

- Enrichment Questions
  - These questions are intended to deepen your understanding.
  - Will *NOT* appear in the exams
  - Participation is optional and for your interest.
  - If you'd like to discuss them, feel free to email me or your tutor (cvičiaci) to arrange a meeting.

Eulerian tours and Hamiltonian cycles

- Definitions and terminologies
- Different types of graphs
- Touring a graph
- Eulerian circuit algorithms
- Eulerization
- Existence of Hamiltonian cycles
- Traveling salesman problem
- Digraphs and Asymmetric Traveling Salesman Problem

Hamiltonian cycle and Hamiltonian path

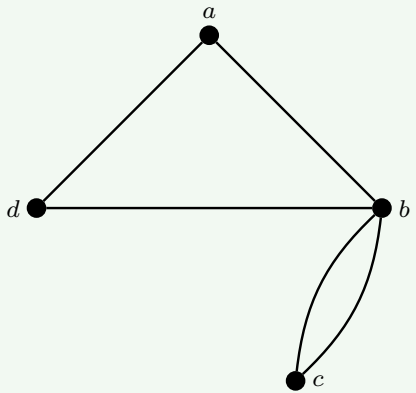
Definition

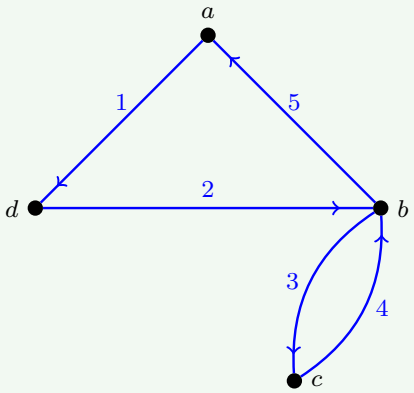
A cycle in a graph G that contains every vertex of G is called a *Hamiltonian cycle*. A path that contains every vertex is called a *Hamiltonian path*.

- Recall that a cycle or a path can only pass through a vertex once, so the Hamiltonian cycles and paths travel through every vertex exactly once.
- Named after mathematician: Sir William Hamilton

Hamiltonian cycle and Eulerian circuit – example

Example

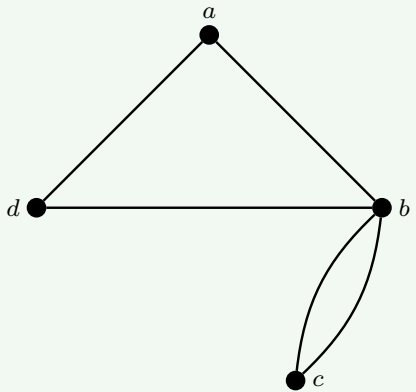


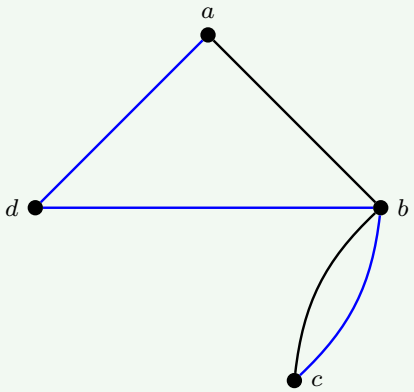


The graph is connected and all vertices are even – it contains an Eulerian circuit

Hamiltonian cycle and Eulerian circuit – example

Example

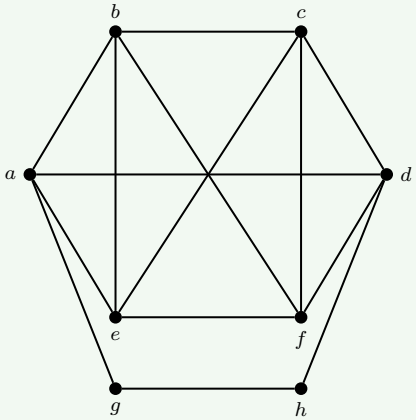


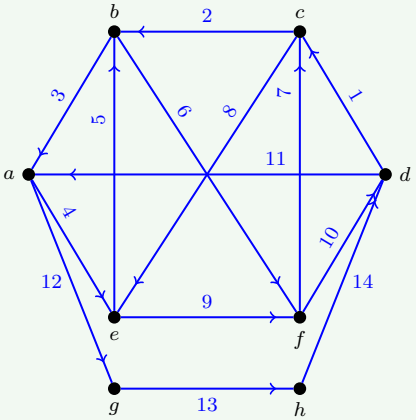


There is no Hamiltonian cycle since we need to include c in the cycle and by doing so we would pass b twice

Hamiltonian cycle and Eulerian circuit – example

Example

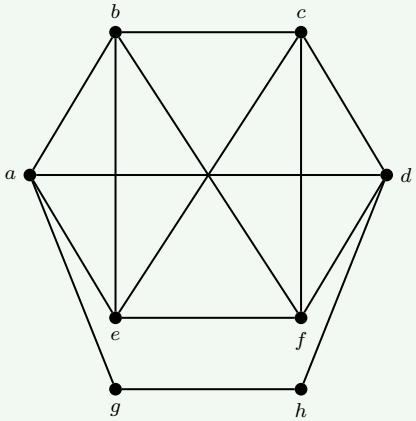


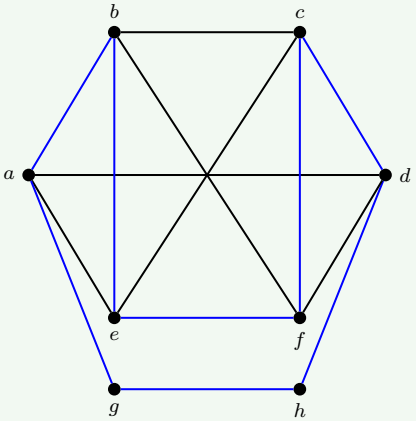


The graph is connected and all vertices are even – it contains an Eulerian circuit

Hamiltonian cycle and Eulerian circuit – example

Example





Hamiltonian cycles and Hamiltonian paths also exist. To find one such path, remove any one of the highlighted edges from the Hamiltonian cycle

Hamiltonian Cycle and Hamiltonian Path

- If a graph has an Eulerian circuit, then it cannot have an Eulerian trail, and vice versa
- This is not true for the Hamiltonian version
  - If a graph has a Hamiltonian cycle, it automatically has a Hamiltonian path (just leave off the last edge of the cycle to obtain a path).
  - If a graph has a Hamiltonian path, it may or may not have a Hamiltonian cycle.

Necessary conditions for Hamiltonian cycle

- G must be connected
- No vertex of G can have degree less than 2
- G cannot contain a cut-vertex

Cut-vertex: a vertex whose removal disconnects the graph

Note

These are not sufficient conditions

A sufficient condition for Hamiltonian cycle

Theorem (Dirac's Theorem)

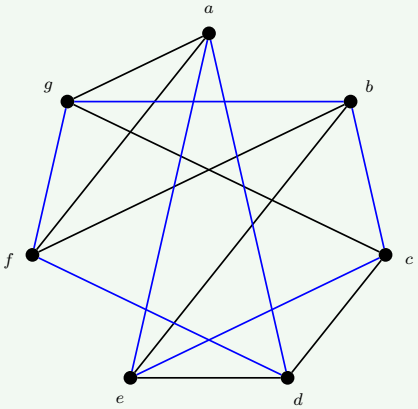
Let G be a graph with $n \geq 3$ vertices. If every vertex v of G satisfies $\deg(v) \geq \frac{n}{2}$, then G has a Hamiltonian cycle.

Remark

- The proof of this theorem boils down to the fact that each vertex has so many edges incident to it that in trying to find a cycle we will never get stuck at a vertex.
- There are several sufficient conditions for a graph to have a Hamiltonian cycle, but there is no known simple necessary and sufficient condition that works for all graphs.

Existence of Hamiltonian cycle – example

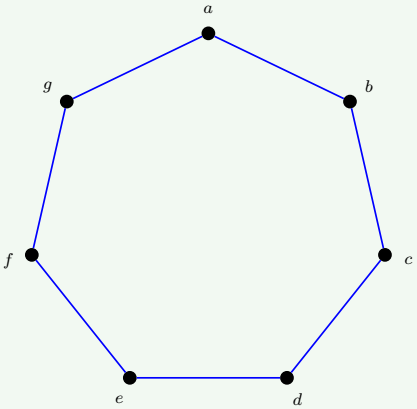
Example



$n = 7$

$\deg = 4 \geq \frac{7}{2}$

for all vertices



$n = 7$

$\deg = 2 < \frac{7}{2}$

for all vertices

Complete graph

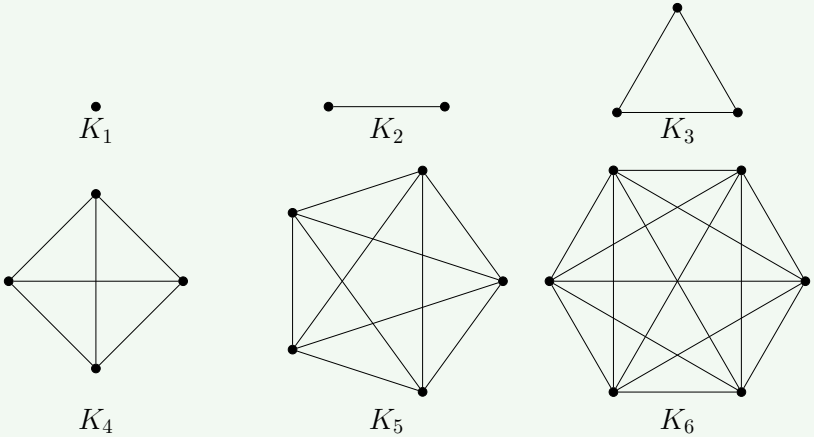
Definition

A simple graph G is *complete* if every pair of distinct vertices is adjacent. The complete graph on n vertices is denoted K_n.

- If we think of an edge as describing a relationship between two objects, then a complete graph represents a scenario where every pair of vertices satisfies this relationship.

The first six complete graphs

Example



Properties of K_n

- Each vertex in K_n has degree $n-1$
- K_n has $\frac{n(n-1)}{2}$ edges
- K_n contains the most edges out of all simple graphs on n vertices

Note

- Any complete graph (on at least 3 vertices) satisfies the conditions of Dirac's Theorem, and therefore contains a Hamiltonian cycle

Number of Hamiltonian cycles in a complete graph

- When modeling a problem whose solution is a Hamiltonian cycle (or path) in the appropriate graph, there is often a desired "home" location or starting vertex – *reference point*

Theorem

Given a specified reference point, the complete graph K_n has $(n-1)!$ distinct Hamiltonian cycles. Half of these cycles are reversals of the others.

- When starting at the designated vertex there are $n-1$ possible edges to choose from
- Once that edge has been traveled to arrive at a new vertex, we cannot pick the edge just traveled and so there remain $n-2$ edges to choose from
- At the next vertex, we cannot travel back to either of the previously chosen vertices, and so there remains $n-3$ edges available
- Continuing in this manner, we have a total of

$$(n-1)(n-2)(n-3)\cdots(2)(1) = (n-1)!$$

Eulerian tours and Hamiltonian cycles

- Definitions and terminologies
- Different types of graphs
- Touring a graph
- Eulerian circuit algorithms
- Eulerization
- Existence of Hamiltonian cycles
- Traveling salesman problem
- Digraphs and Asymmetric Traveling Salesman Problem

The question

- How should a delivery service plan its route through a city to ensure each customer is reached?
- A traveling salesman has customers in numerous cities. He must visit each of them and return home, but wishes to do this with the least total cost
- Shortest Hamiltonian cycle visiting all cities in one country
- Model: weighted complete graph
- Weights: different distances
- Complete: it is reasonable to assume it's possible to travel between any two cities

Brute Force Algorithm

- To find the Hamiltonian cycle of least total weight, one obvious method is to find all possible Hamiltonian cycles and pick the cycle with the smallest total.
- The method of trying every possibility to find an optimal solution is referred to as an *exhaustive search*, or use of the *Brute Force Algorithm*.
- Input: weighted complete graph K_n
- Output: Minimum Hamiltonian cycle

Brute Force Algorithm

1. Choose a starting vertex, call it v
2. Find all Hamiltonian cycles starting at v. Calculate the total weight of each cycle.
3. Compare all $(n-1)!$ cycles. Pick one with the least total weight.

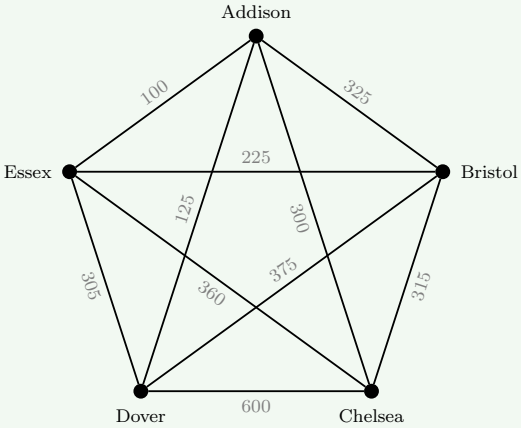
Note

In step 3, there should be at least two options

Brute Force Algorithm – example

Example

Liz is planning her next business trip from her hometown of Addison and has determined the cost for travel between any of the five cities she must visit.



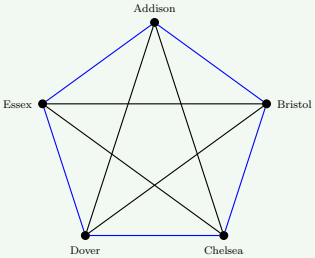
Brute Force Algorithm – example

Example

- Number of tours starting and ending at Addison: $4! = 24$.
- These represent 12 distinct Hamiltonian cycles, since each cycle is counted twice by reversal.
- One method for finding all Hamiltonian cycles, and ensuring you indeed have all 24, is to use alphabetical or lexicographic ordering of the cycles.
- Note that all cycles must start and end at Addison, and we will abbreviate all cities with their first letter
- For example, the first cycle is $abcdea$

Brute Force Algorithm – example

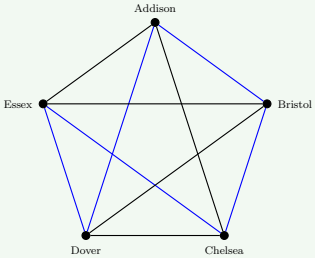
Example



$abcdea$

$aedcba$

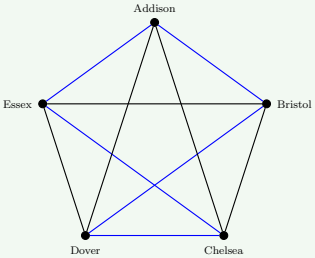
1645



$abceda$

$adecba$

1430



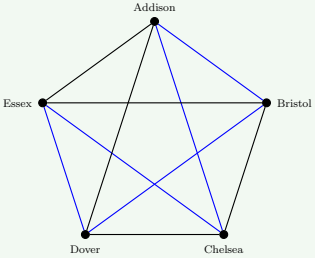
$abdcea$

$aecdba$

1760

Brute Force Algorithm – example

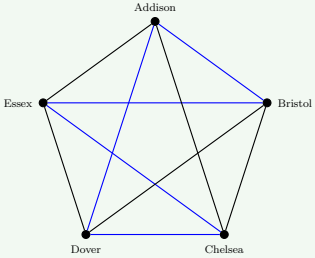
Example



$abdeca$

$acedba$

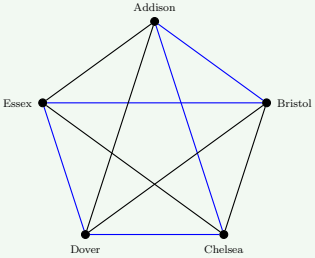
1665



$abecda$

$adceba$

1635



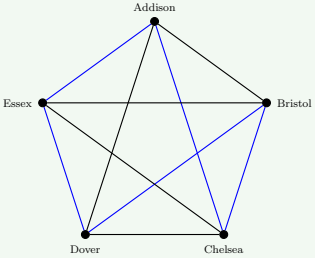
$abedca$

$acdeba$

1755

Brute Force Algorithm – example

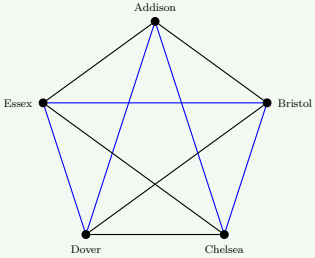
Example



$acbdea$

$aedbca$

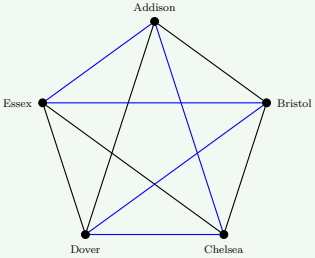
1395



$\boldsymbol{acbeda}$

$\boldsymbol{adebca}$

1270



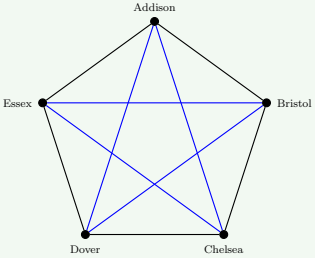
$acdbea$

$aebdca$

1600

Brute Force Algorithm – example

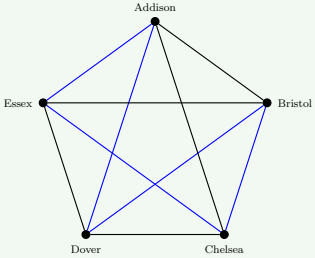
Example



$acebda$

$adbeca$

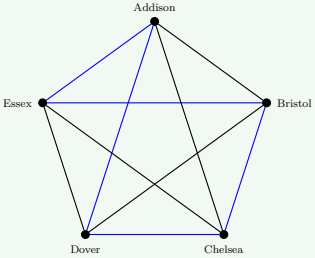
1385



$adbcea$

$aecbda$

1275



$adcbea$

$aebcda$

1365

Estimated effort

- Effort for using Brute Force Algorithm to find the optimal Hamiltonian cycle in K_n
- Best: best supercomputer
- Top 500: top 500 supercomputers
- Earth is about 4.54×10^9 years old

	Supercomputers	
n	**Best**	**Top 500**
5	2×10^{-15} seconds	2×10^{-16} seconds
15	2×10^{-5} seconds	2×10^{-6} seconds
20	40 seconds	4 seconds
21	14 minutes	2 minutes
22	5 hours	32 minutes
23	4.5 days	12 hours
24	16 weeks	12 days
25	7.5 years	10 months
26	2 centuries	2 decades
30	132 million years	14 million years
40	4.1×10^{23} years	4.3×10^{22} years
50	1.4×10^{40} years	1.5×10^{39} years

Other algorithms

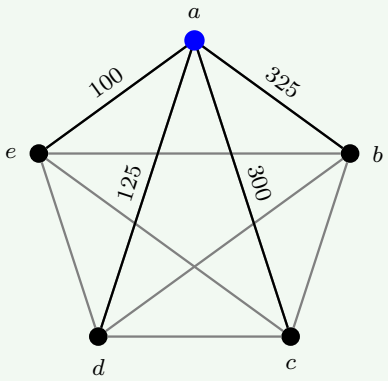
- We will discuss a few more efficient algorithms
- Mathematicians have been searching for algorithms that will find the optimal cycle in a relatively short time span; that is, an algorithm that is both efficient and optimal
- Not only has no such algorithm been found for the Traveling Salesman Problem, but some mathematicians believe no such algorithm even exists.

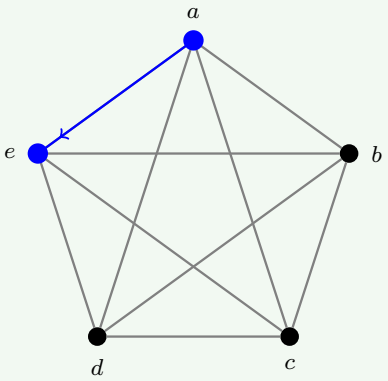
Nearest Neighbor Algorithm

1. Choose a starting vertex, say v. Highlight v
2. Among all edges incident to v, pick the one with the smallest weight. If more than one possible choices have the same weight, randomly pick one.
3. Highlight the edge and move to its other endpoint u. Highlight u
4. Repeat steps 2 and 3, where only edges to unhighlighted vertices are considered
5. Close the cycle by adding the edge to v from the last vertex highlighted. Calculate the total weight.

Nearest Neighbor Algorithm – example

Example

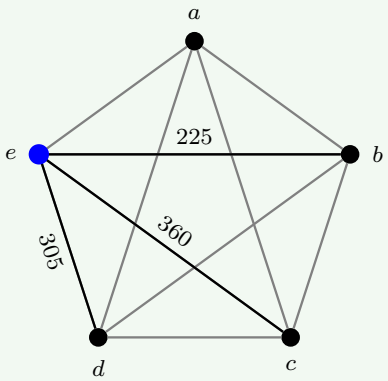


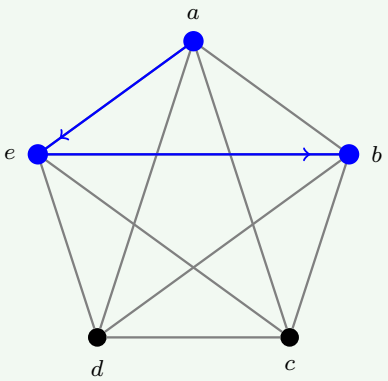


- Step 1: Starting vertex is a
- Step 2: Edge of smallest weight: ae
- Step 3: Move to vertex e. Highlight e

Nearest Neighbor Algorithm – example

Example

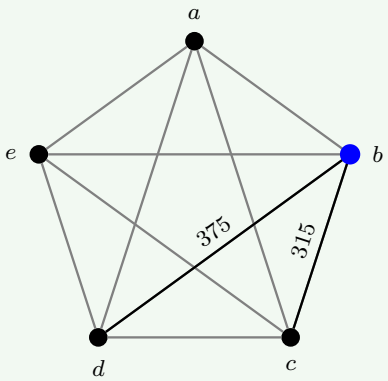


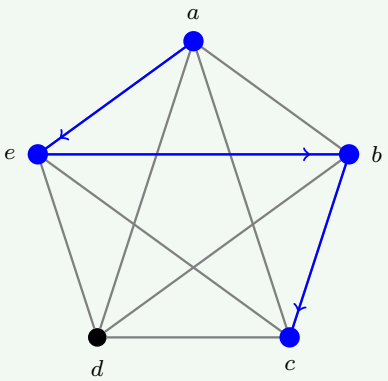


- Step 2: Edge of smallest weight: eb, note here we do not consider edge ea
- Step 3: Move to vertex b. Highlight b

Nearest Neighbor Algorithm – example

Example

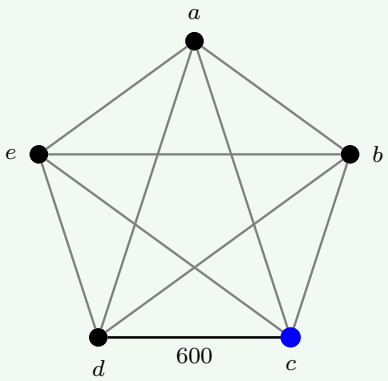


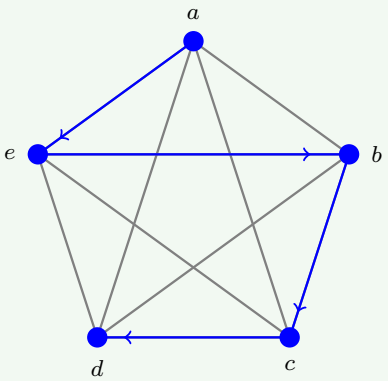


- Step 2: Edge of smallest weight: bc
- Step 3: Move to vertex c. Highlight c

Nearest Neighbor Algorithm – example

Example

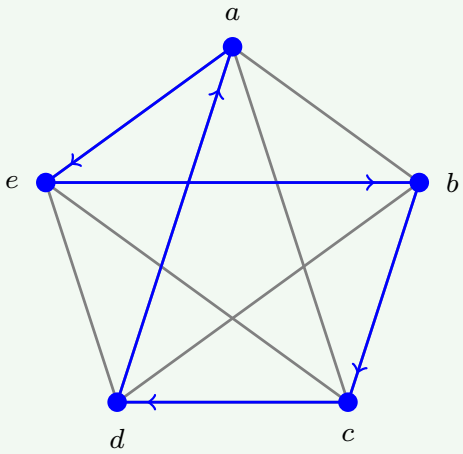




- Step 2: Only choice of edge: cd
- Step 3: Move to vertex d. Highlight d

Nearest Neighbor Algorithm – example

Example



- Step 5: close the cycle. Total weight is 1365

Remarks

- The last two edges are completely determined since we cannot travel back to vertices that have already been chosen
  - This could force us to use the heaviest edges in the graph, as happened above
- The arbitrary choice of a starting vertex could cause light edges to be eliminated from consideration

Though we cannot do anything about the former concern, we can address the latter – by using a different starting vertex, the Nearest Neighbor Algorithm may identify a new Hamiltonian cycle, which may be better or worse than the initial cycle.

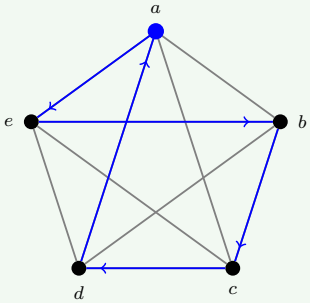
Repetitive Nearest Neighbor Algorithm

1. Choose a starting vertex, say v
2. Apply the Nearest Neighbor Algorithm
3. Repeat steps 1 and 2 so each vertex of K_n serves as the starting vertex
4. Choose the cycle of least total weight. Rewrite it with the desired reference point.

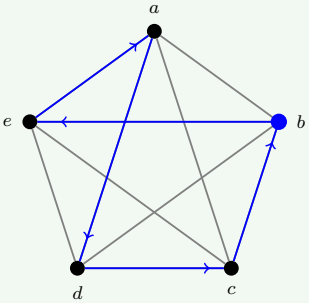
Repetitive Nearest Neighbor Algorithm – example

Example

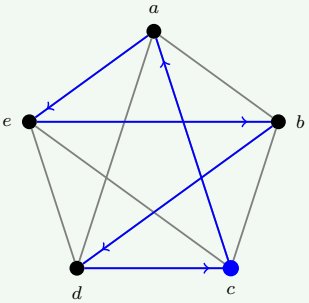
With the same example as before, we can obtain the following cycles with the Repetitive Nearest Neighbor Algorithm



$aebcda$
$aebcda$
1365



$beadcb$
$adcbea$
1365

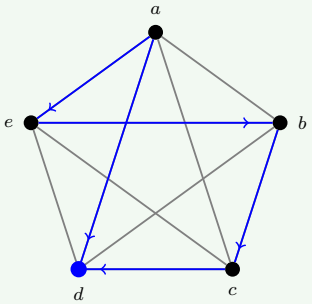


$caebdc$
$aebdca$
1600

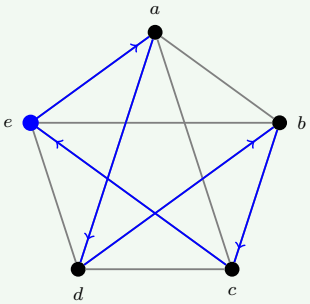
Repetitive Nearest Neighbor Algorithm – example

Example

With the same example as before, we can obtain the following cycles with the Repetitive Nearest Neighbor Algorithm



$daebcd$
$aebcda$
1365



$eadbce$
$adbcea$
1275

Remarks

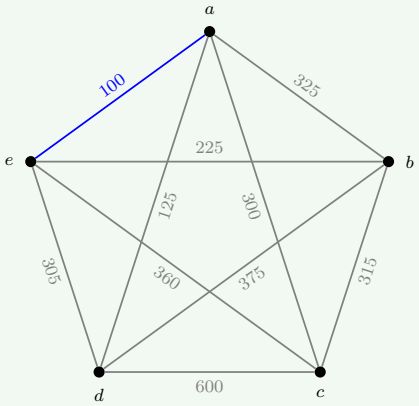
- It should come as no surprise that Repetitive Nearest Neighbor performs better than Nearest Neighbor; however, there is no guarantee that this improvement will produce the optimal cycle.
- Even though Repetitive Nearest Neighbor addressed the concern of missing small weight edges, it is still possible that some of these will be bypassed as we travel a cycle
- Cheapest Link Algorithm: choosing edges in order of weight as opposed to edges along a tour

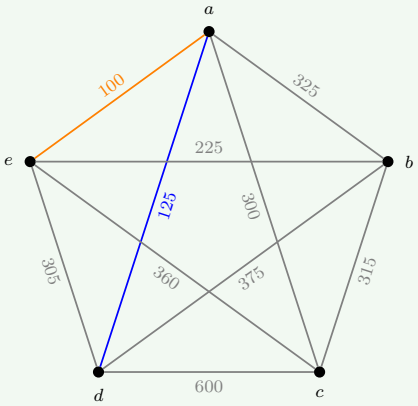
Cheapest Link Algorithm

1. Among all edges in the graph, pick the one with the smallest weight. If more than one possible choices have the same weight, randomly choose one. Highlight the chosen edge
2. Repeat step 1 with the added conditions
   - no vertex has three highlighted edges incident to it
   - no edge is chosen if it would create a cycle that does not include all vertices
3. Calculate the total weight

Cheapest Link Algorithm – example

Example

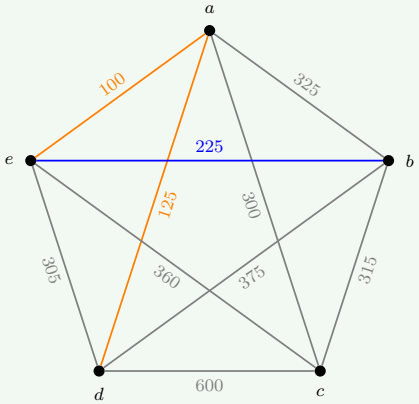


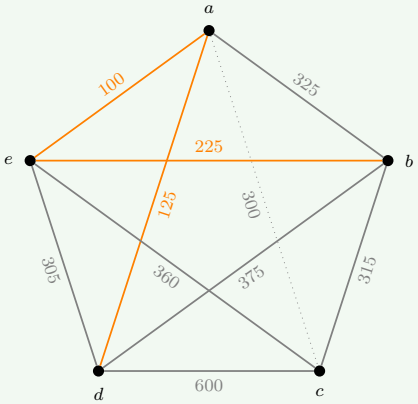


- Step 1. The smallest weight is 100 for edge ae
- Step 2. The next smallest weight is 125 with edge ad

Cheapest Link Algorithm – example

Example

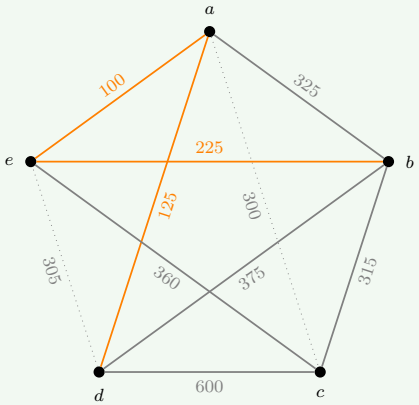


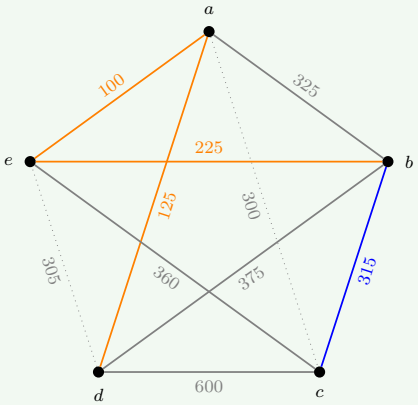


- Step 2. The next smallest weight is 225 for be
- Step 2. ac has the next smallest weight, but highlighting ac will cause a to have three incident edges

Cheapest Link Algorithm – example

Example

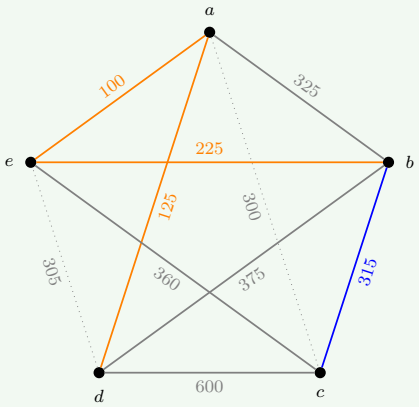


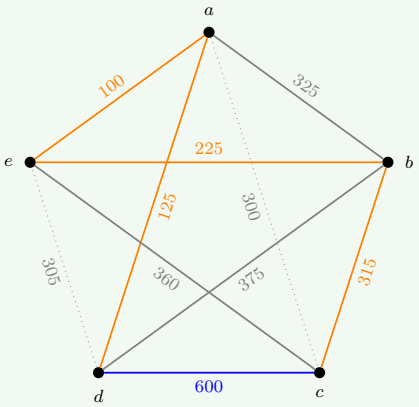


- Step 2. ed has the next smallest weight, but highlighting ed will create a cycle $eade$ that does not contain all vertices, also give e three incident highlighted edges
- Step 2. next available is bc

Cheapest Link Algorithm – example

Example





- Step 2. at this point we must close the cycle and there is only one choice cd
- Output: resulting cycle is $aebcda$ with total weight 1365

Remarks

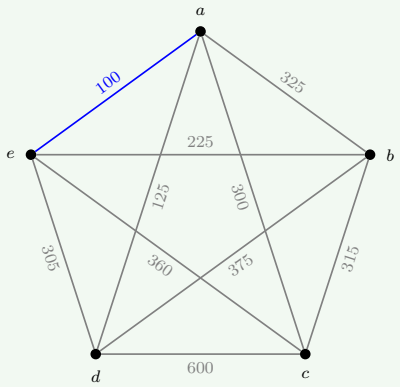
- In the example above, Cheapest Link ran into the same trouble as the initial cycle created using Nearest Neighbor
- Although the lightest edges were chosen, the heaviest also had to be included due to the outcome of the previous steps
- The downfall of the last two algorithms is the focus on choosing the smallest weighted edges at every opportunity.
- In some instances, it may be beneficial to work for a balance —choose more moderate weight edges to avoid later using the heaviest.
- Nearest Insertion Algorithm: forms a small circuit initially from the smallest weighted edge and balances newly added edges with the removal of a previously chosen edge.

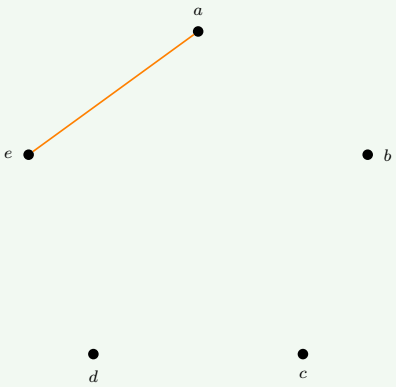
Nearest Insertion Algorithm

1. Among all edges in the graph, pick one with the smallest weight. Highlight the edge and its endpoints.
2. Pick a vertex that is closest to one of the two already chosen vertices. Highlight the new vertex and its edges to both of the previously chosen vertices.
3. Identify the unvisited vertex that is closest to any vertex already in the current cycle. Insert this vertex into the existing cycle by connecting it to the nearest chosen vertex. Then, add a second edge to complete the insertion and remove one existing edge to maintain a cycle. Choose the scenario with the smallest total weight.
4. Repeat step 3 until all vertices have been included in the cycle
5. Calculate the total weight

Nearest Insertion Algorithm – example

Example

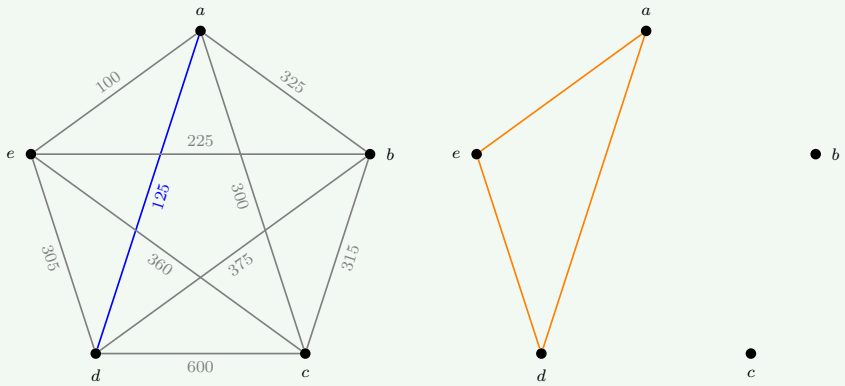




- Step 1. the smallest weight edge is ae

Nearest Insertion Algorithm – example

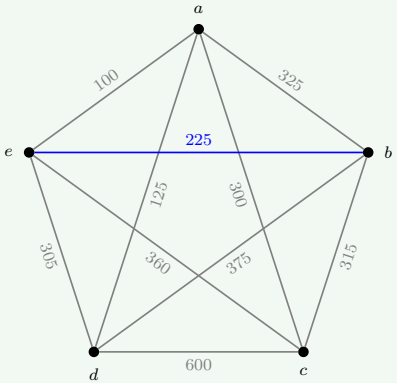
Example

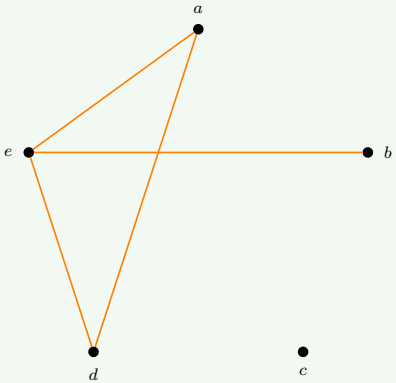


- Step 2: closest vertex to either a or e is d through edge ad. Form a cycle by adding ad and de
- Step 3: closes vertex to any of a, d or e is b through the edge be

Nearest Insertion Algorithm – example

Example



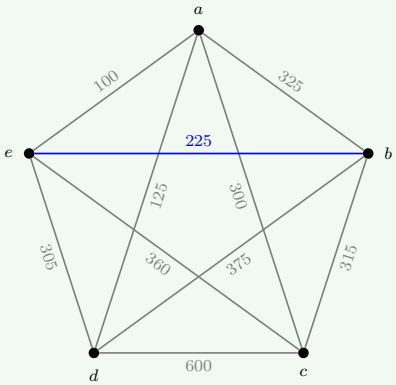


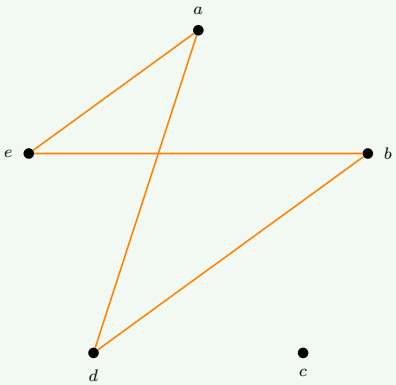
Step 3: After adding edge be, we need to decide whether to add ba and delete ea or add bd and delete ed

$$\begin{aligned} \omega(ba) - \omega(ea) &= 325 - 100 = 225 \\ \omega(bd) - \omega(ed) &= 375 - 305 = \mathbf{70} \end{aligned}$$

Nearest Insertion Algorithm – example

Example

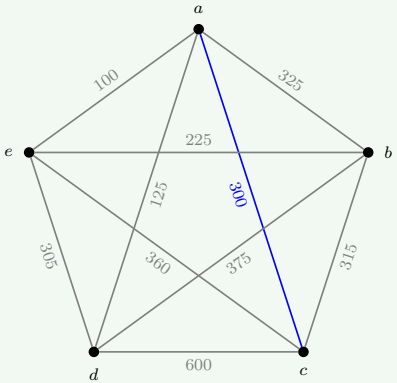


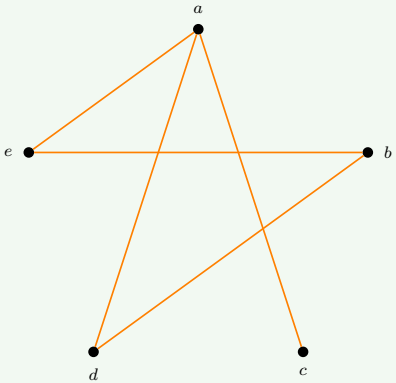


- Step 3: We add edge bd and remove edge ed to get a bigger cycle than before
- Step 3: Only vertex left is c, c is closest to a

Nearest Insertion Algorithm – example

Example



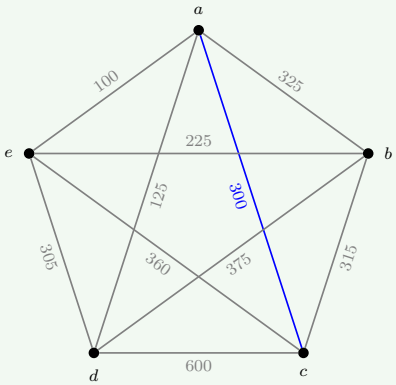


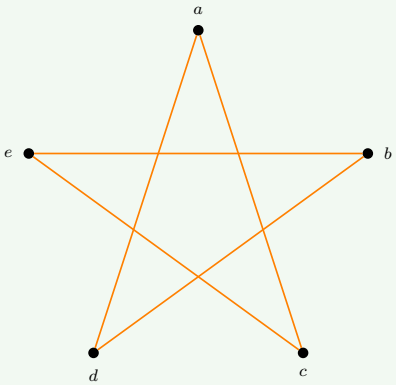
Step 3: After adding edge ca, we need to decide whether to add cd and delete ad or add ce and delete ae

$$\begin{aligned} \omega(cd) - \omega(ad) &= 600 - 125 = 475 \\ \omega(ce) - \omega(ae) &= 360 - 100 = \mathbf{260} \end{aligned}$$

Nearest Insertion Algorithm – example

Example





- Step 3: We add edge ce and remove edge ae to get a bigger cycle than before
- Step 5: The cycle is $acebda$ with total weight 1385.

Eulerian tours and Hamiltonian cycles

- Definitions and terminologies
- Different types of graphs
- Touring a graph
- Eulerian circuit algorithms
- Eulerization
- Existence of Hamiltonian cycles
- Traveling salesman problem
- Digraphs and Asymmetric Traveling Salesman Problem

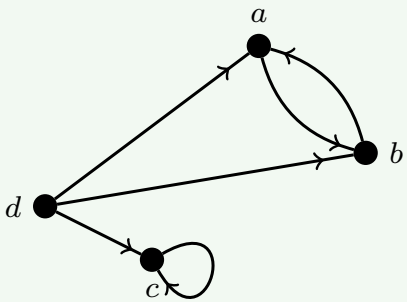
Directed graph

Definition

A *directed graph*, or *digraph*, is a graph $G = (V, A)$ that consists of a vertex set V and an *arc set* A. An *arc* is an ordered pair of vertices.

Example

Let G be a digraph with $V = \{a, b, c, d\}$ and $A = \{ab, ba, cc, dc, db, da\}$.



Terminologies

Definition

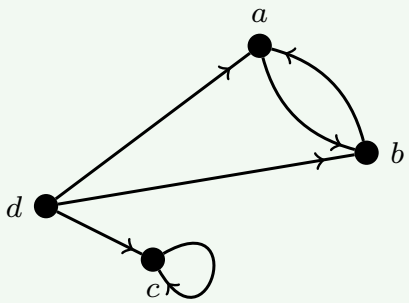
Let $G = (V, A)$ be a digraph

- Arc yx, x is *head*, y is *tail*
- *in-degree* of vertex x: the number of arcs for which x is a head, denoted $\deg^-(x)$
- *out-degree* of vertex x: the number of arcs for which x is the tail, denoted $\deg^+(x)$
- The *underlying graph* for a digraph is the graph $G' = (V, E)$ which is formed by removing the direction from each arc to form an edge
- A *directed path* is a path in which the head of an arc is the tail of the next arc in the path
- A *directed cycle* is a cycle in which the head of an arc is the tail of the next arc in the cycle

Terminologies – example

Example

- a is the tail of arc ab and the head of arcs da and ba
- $\deg^-(a) = 2$, $\deg^-(b) = 2$, $\deg^-(c) = 2$, $\deg^-(d) = 0$
- $\deg^+(a) = 1$, $\deg^+(b) = 1$, $\deg^+(c) = 1$, $\deg^+(d) = 3$
- $d \to a \to b$ is a directed path
- $c \to d \to a \to b$ is not a directed path since cd is not an arc in the graph
- $a \to b \to a$ is a directed cycle
- $d \to a \to b \to d$ is not a directed cycle



Degrees of vertices

Theorem

Let $G = (V, A)$ be a digraph and $|A|$ denote the number of arcs in G. Then both the sum of the in-degrees of the vertices and the sum of the out-degrees equals the number of arcs. In other words, if $V = \{v_1, v_2, \ldots, v_n\}$, then

$$\deg^-(v_1) + \cdots \deg^-(v_n) = |A| = \deg^+(v_1) + \cdots \deg^+(v_n)$$

- Since each arc will contribute to the in-degree to its head and to the out-degree of its tail, the sum of the in-degrees equals the sum of the out-degrees.
- Each arc is counted exactly once in the sum of the in-degrees (or out-degrees) since it has a unique head (or tail).

Hamiltonian cycles

A few properties should be immediately apparent for a digraph to have a Hamiltonian cycle

- G must be connected
- No vertex of G can have in-degree or out-degree 0
- G cannot contain a cut-vertex

A sufficient condition for Hamiltonian cycles

Theorem

Let G be a digraph with $n \geq 3$ vertices. If

$$\deg^{-}(v) \geq \frac{n}{2}, \quad \deg^{+}(v) \geq \frac{n}{2}$$

for every vertex v of G, then G has a Hamiltonian cycle.

Asymmetric Traveling Salesman Problem

- The Asymmetric Traveling Salesman Problem is to find the optimal Hamiltonian *directed* cycle on a digraph.
- The weight of the arc ab need not equal the weight of arc ba
- *Complete digraph*: for each pair of vertices x and y both arcs xy and yx exist in the graph
- We will only consider complete digraphs in which each arc has a positive weight
- To utilize the algorithms discussed before, we convert a digraph into an undirected graph

Undirecting Algorithm

Input: Weighted complete digraph $G = (V, A, \omega)$

Steps:

1. For each vertex x, make a clone x'. Form the edge xx' with weight 0
2. For each arc xy form the edge $x'y$
3. The weight of an edge is equal to the weight of its corresponding arc
   - $\omega(xx') = 0$
   - $\omega(x'y) = \omega(xy)$
   - $\omega(xy') = \omega(yx)$

Output: weighted clone graph $G' = (V', E, \omega')$

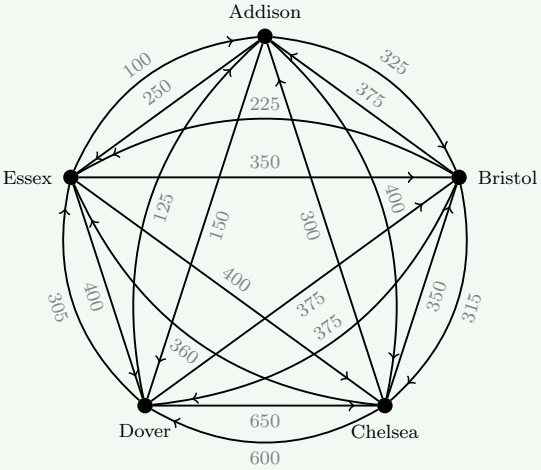
Note

- We need to ensure that when forming a cycle an arc into a vertex is immediately followed by an arc out of that same vertex
- By giving the edge between a vertex and its clone a weight of 0 (when all other arcs have positive weights), we ensure this edge will always be included in any Hamiltonian cycle.
- In addition, this extra edge does not impact the total weight of the final cycle

Undirecting Algorithm – example

Example

Liz is heading out on the road to visit her customers, the direction of a route impacts its cost



Undirecting Algorithm – example

Example

Table representation. e.g. Addison to Bristol - 325

	Addison	Bristol	Chelsea	Dover	Essex
Addison	.	325	400	150	250
Bristol	375	.	315	375	225
Chelsea	300	350	.	600	360
Dover	125	375	650	.	305
Essex	100	350	400	400	.

Undirecting Algorithm – example

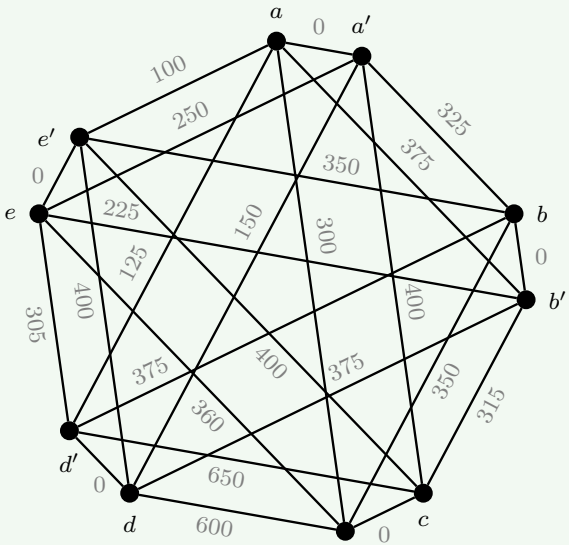
Example

Note: original edges do not exist. The table is symmetric, with a copy of the original table in the lower left quadrant and its mirror image in the upper right quadrant.

	a	b	c	d	e	a'	b'	c'	d'	e'
a	.	325	400	150	250	0	375	300	125	100
b	375	.	315	375	225	325	0	350	375	350
c	300	350	.	600	360	400	315	0	650	400
d	125	375	650	0	305	150	375	600	0	400
e	100	350	400	400	0	250	225	360	305	0
a'	0	325	400	150	250	.	.	.	.	.
b'	375	0	315	375	225	.	.	.	.	.
c'	300	350	0	600	360	.	.	.	.	.
d'	125	375	650	0	305	.	.	.	.	.
e'	100	350	400	400	0	.	.	.	.	.

Undirecting Algorithm – example

Example



Remarks

- Even though the input of the algorithm is a complete digraph, the output is not a complete graph
- The Undirecting Algorithm can be applied to a digraph that is not complete; however, the resulting graph may not have a Hamiltonian cycle.
- When applied to a complete digraph G with n vertices, the resulting graph G' has $2n$ vertices, each of degree n, is guaranteed to have a Hamiltonian cycle.
- The cycle from G' can then be translated to a Hamiltonian directed cycle of the digraph G where vertex copies get reduced back to a single vertex

Asymmetric Traveling Salesman Problem – example

Example

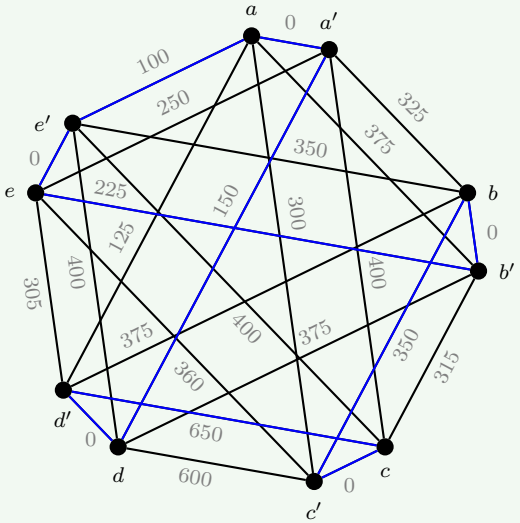
Continue from the previous example, the table below lists the four cycles found using Nearest Neighbor and their conversion to a directed cycle in the digraph.

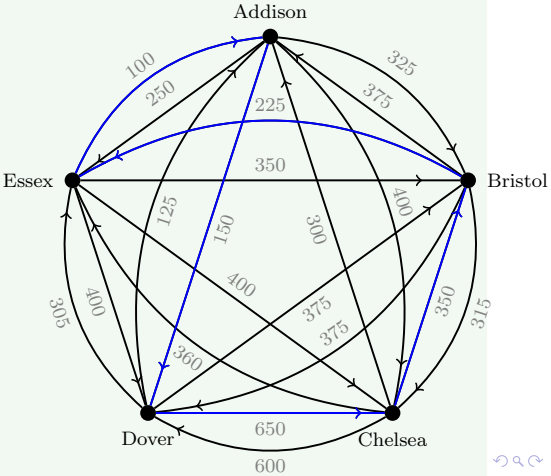
Nearest Neighbor Cycle	Conversion	Total Weight
$a\ a'\ d\ d'\ e\ e'\ b\ b'\ c\ c'\ a$	$a \to d \to e \to b \to c \to a$	1420
$a'\ a\ e'\ e\ b'\ b\ c'\ c\ d'\ d\ a'$	$a \to d \to c \to b \to e \to a$	1475
$e\ e'\ a\ a'\ d\ d'\ b\ b'\ c\ c'\ e$	$e \to a \to d \to b \to c \to e$	1300
$e'\ e\ b'\ b\ c'\ c\ a'\ a\ d'\ d\ e'$	$e \to d \to a \to c \to b \to e$	1500

Note that cycles beginning with a clone vertex must be reversed in the translation back into a direct cycle.

Asymmetric Traveling Salesman Problem – example

Example





Remarks

- Unlike in the undirected case, reversals of a Hamiltonian cycle in a digraph (that is not a complete digraph) might not exist, and those that do will most likely result in a different total weight.
- Thus when applying the Nearest Neighbor Algorithm to graphs formed using the Undirecting Algorithms, we must consider starting at both copies of a vertex, such as a and a', e and e' as in the example

Cheapest Link Algorithm

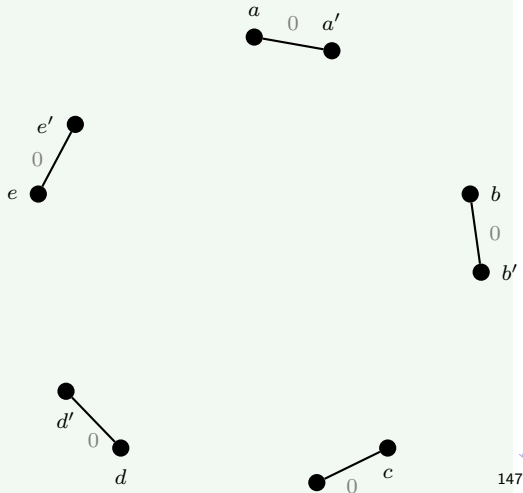
- All edges of weight 0 must be included in the final cycle since picking the edges of minimum weight would initially result in choosing all the weight 0 edges as no two of these are adjacent (so there is no concern of a vertex having degree 3 or closing the circuit too early)

Cheapest Link Algorithm – example

Example

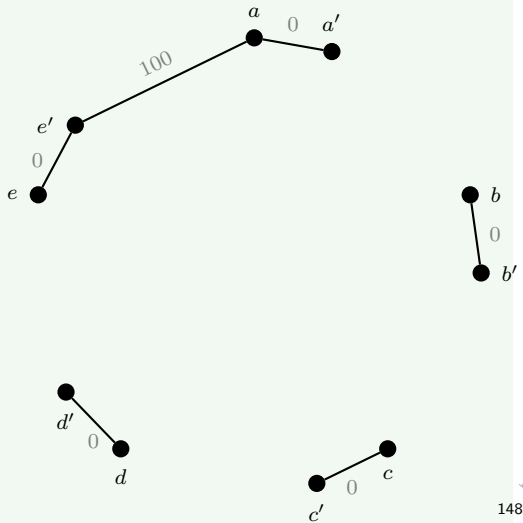
Continuing from the previous example.

- Step 1. Since all edges between a vertex and its clone have a weight of 0, all of these edges will be chosen



Cheapest Link Algorithm – example

Example

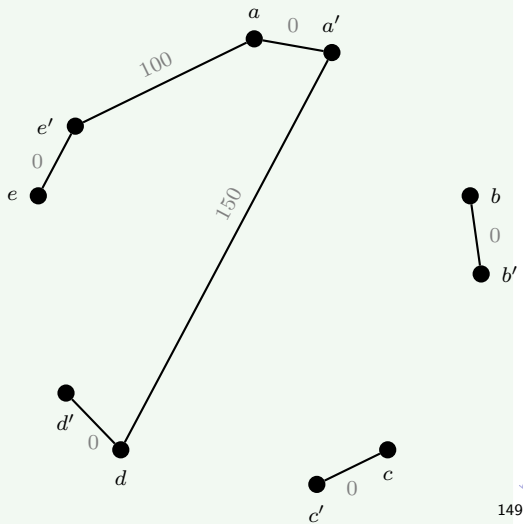


- Step 2. The edge of smallest weight is ae'

Cheapest Link Algorithm – example

Example

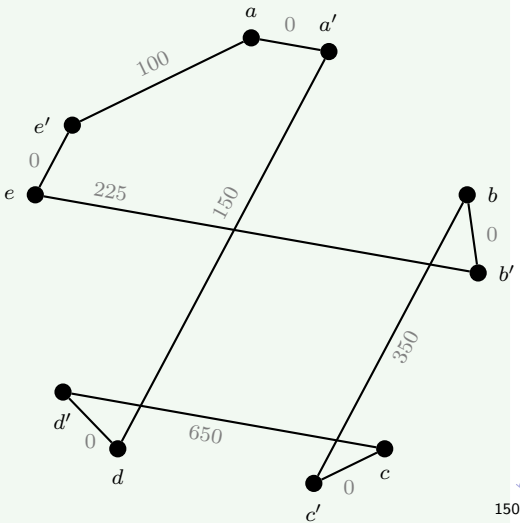
- Step 2. The next lowest is ad'. However, we cannot choose this edge since it would cause a to be incident to three highlighted edges. We skip this edge and move to $a'd$.



Cheapest Link Algorithm – example

Example

- Step 2. Continuing with the Algorithm, we add edges $b'e$, $c'b$. Then we close the cycle with edge cd'.

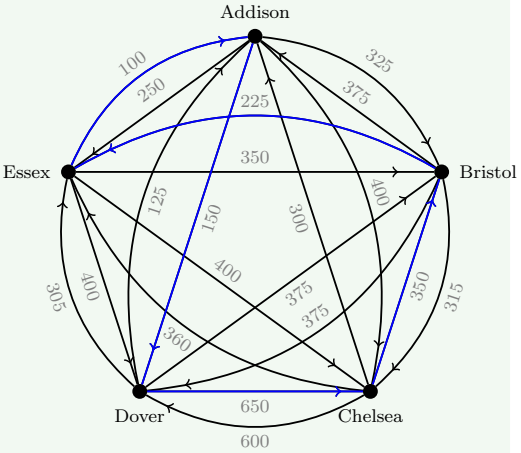


Cheapest Link Algorithm – example

Example

- Step 2. Convert the cycle $aa'dd'cc'bb'ee'a$ to directed cycle in the digraph

$$a \to d \to c \to b \to e \to a$$



Nearest Insertion Algorithm

- The Nearest Insertion Algorithm in its original form does not function on the weighted clone graphs
- There are no cycles of length 3 in the graph created by the undirecting Algorithm and the second step of Nearest Insertion results in a cycle on three vertices originating from the cheapest edge of the graph
- A modification of Nearest Insertion for the weighted clone graphs, which treats the weight 0 edges differently will be discussed during the tutorial